INSTITUTO DE CIENCIAS DE LA EDUCACIÓN · UPM

Javier Galeano · Grupo de Sistemas Complejos · UPM

Machine Learning en el ICE

Clasificación · Regresión · Clustering, en vivo con Orange3





1. Saludo inicial.
2. Presentación.
3. Señalar un vínculo con el interlocutor.
4. Exponer las expectativas.

Como iniciar una conversación de manera correcta

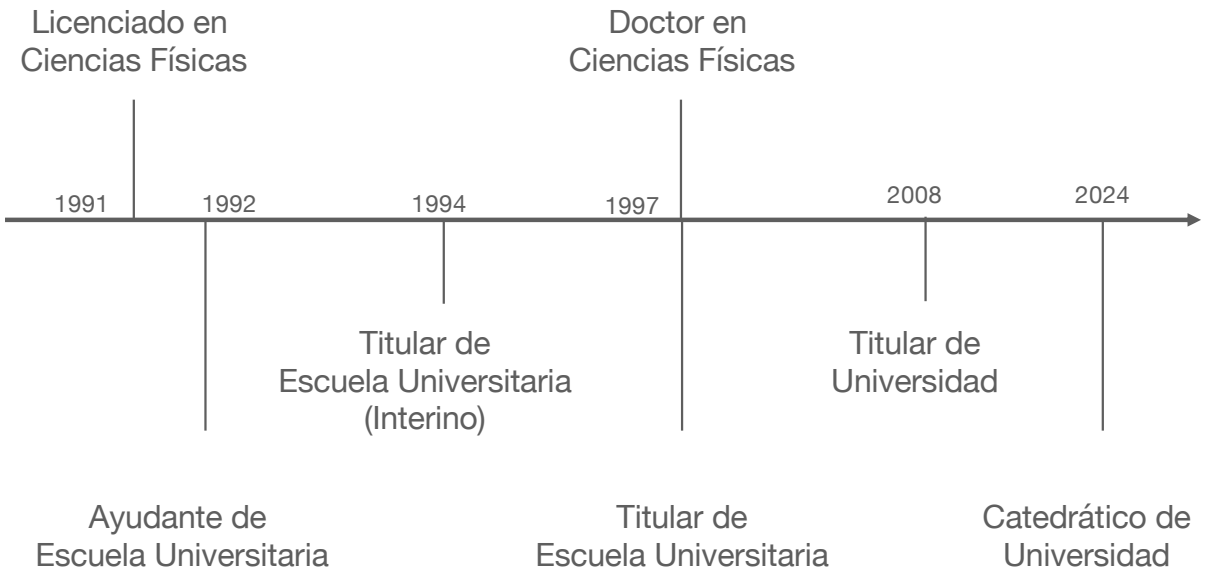


Hola.
Me llamo Javier Galeano.
Y vosotros habéis decidido estudiar ML.
Preparaos a aprender.

(con disculpas a Íñigo Montoya)

Títulos Académicos

Puestos Docentes



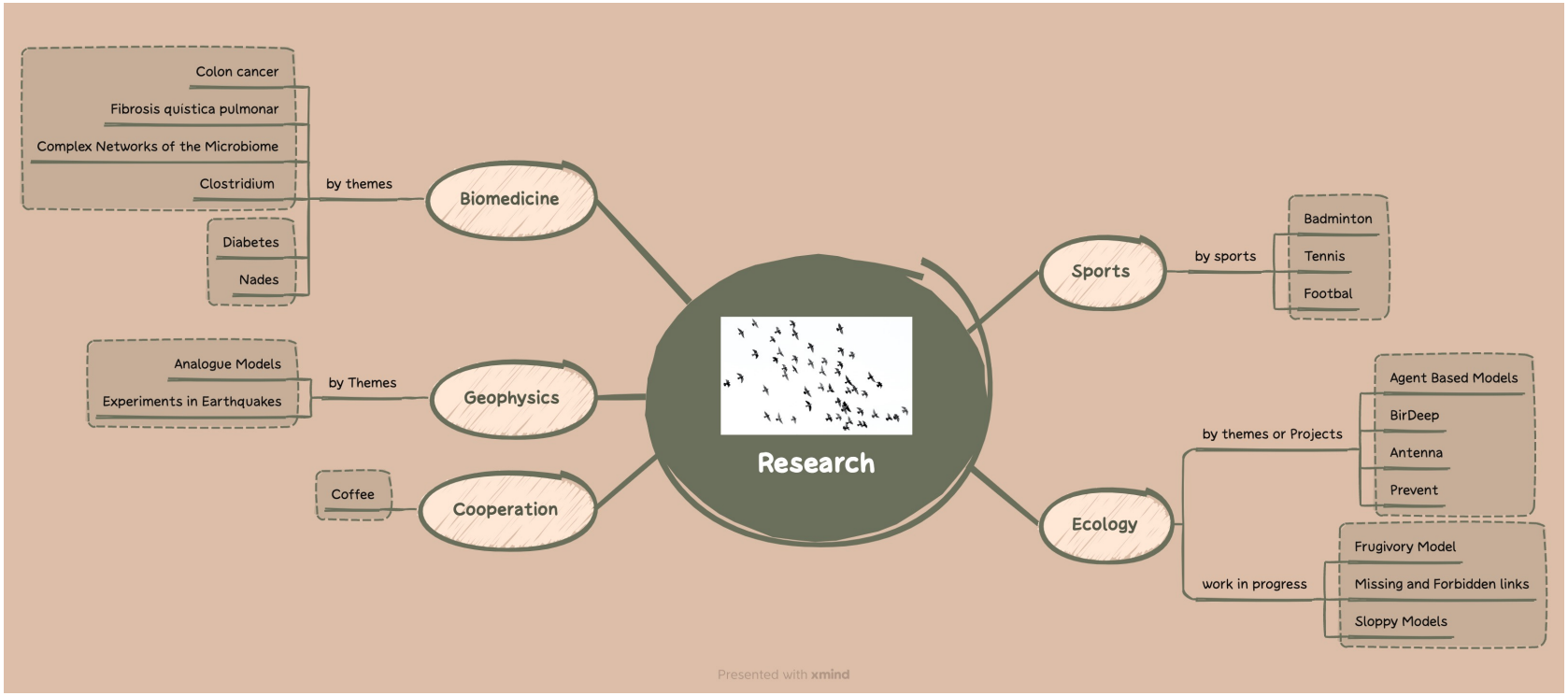
UNIVERSIDAD POLITÉCNICA DE MADRID

Estancias en el Extranjero



Total	Tipo	Semanas	Países
11	6	132	8

Mapa Mental Investigación





Un podcast de culto

S02E03 Un Episodio de mierda: microbios y microbiota. Invitada: Rosa del Campo

orangedatamining.com

Webmail UPM Per... | Diccionarios de E... | Google Maps | YouTube | Wikipedia | Noticias | Populares | Importados de S... | Importados de S... | Todos

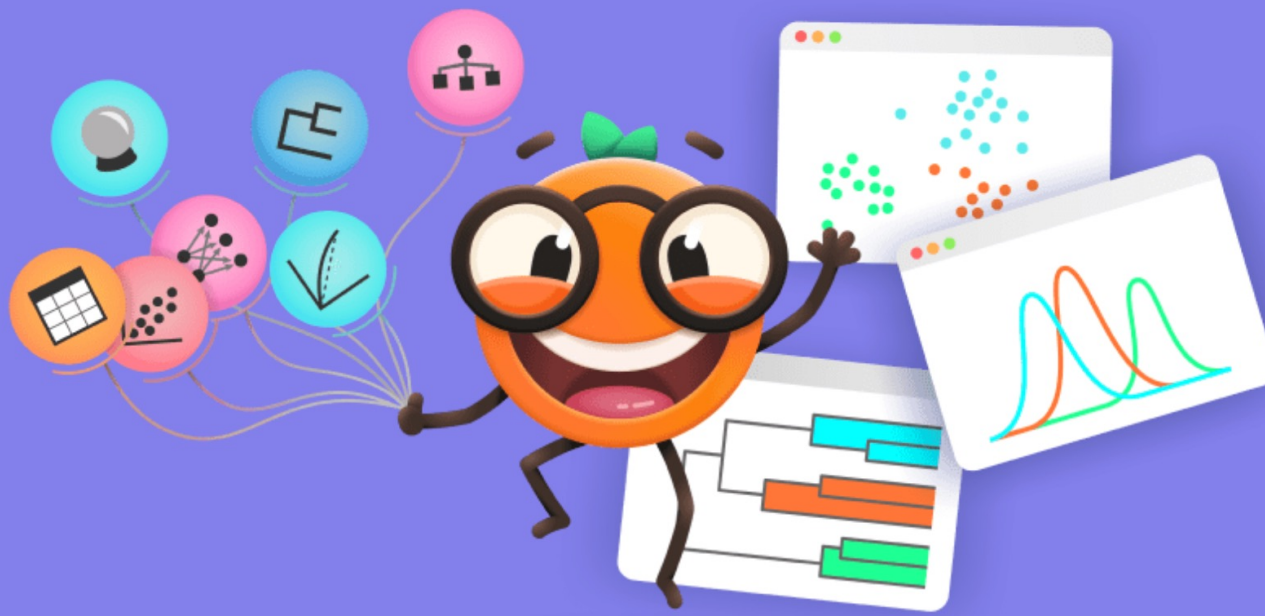
orange
DATA MINING

Examples Download Blog Docs Workshops Search

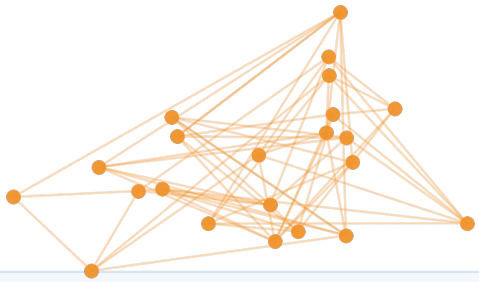
Data Mining Fruitful and Fun

Open source machine learning and data visualization.

Download Orange 3.36.1



Elige tu dataset para el proyecto



Dropout y éxito académico

4424 estudiantes · clasificación (3 clases)
y regresión, con variables socioeconómicas

Rendimiento en exámenes

1000 estudiantes · dataset ligero para ejemplos y pruebas rápidas

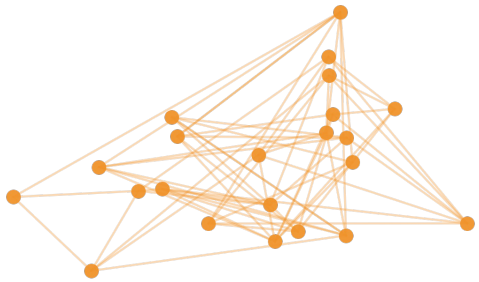
Trae tu propio dataset

Cualquier .csv con tus propios datos,
lo adaptamos juntos a Orange3

Documentación, datasets y guías del curso:

github.com/galeanojav/Curso_ML

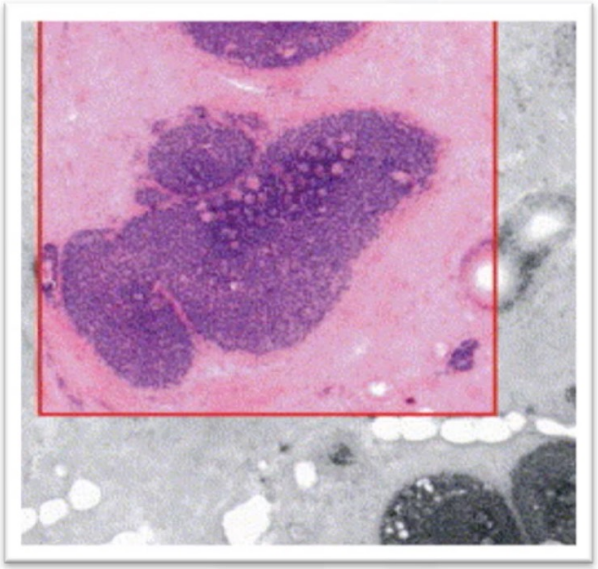
Elige tu dataset para el proyecto

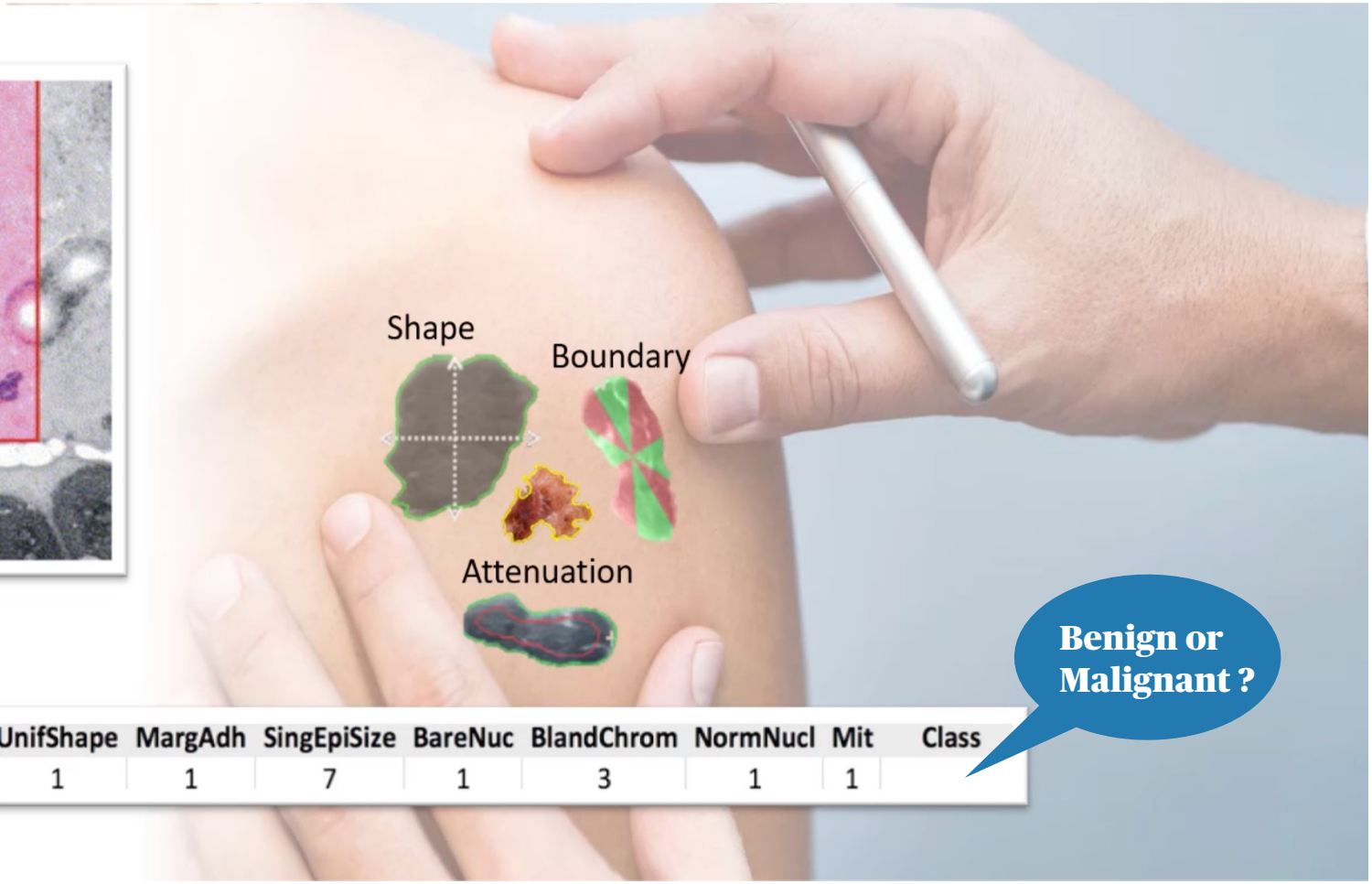


Documentación, datasets y guías del curso:

github.com/galeanojav/Curso_ML

Skin Cancer

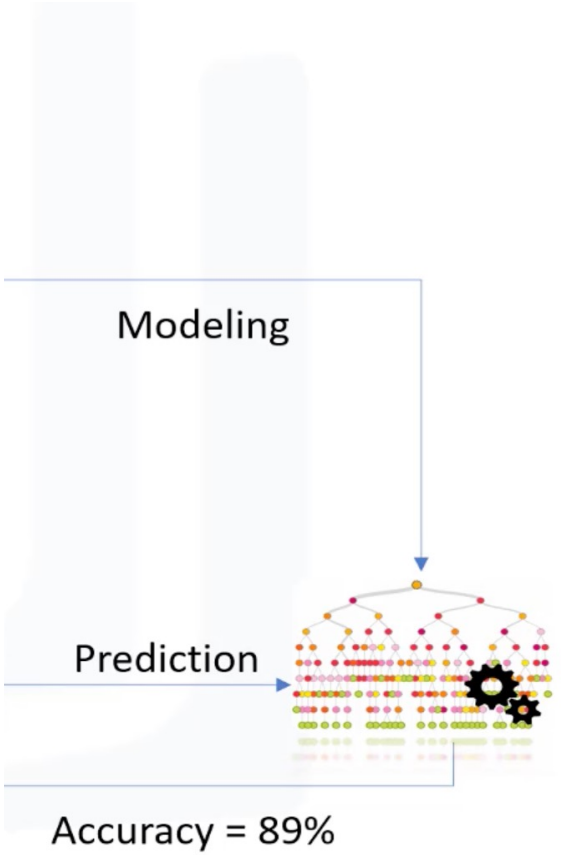




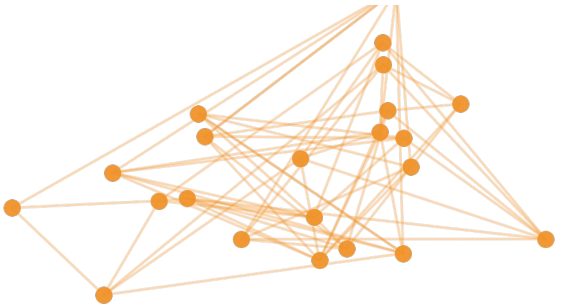
ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000015	6	1	1	1	7	1	3	1	1	

dataset

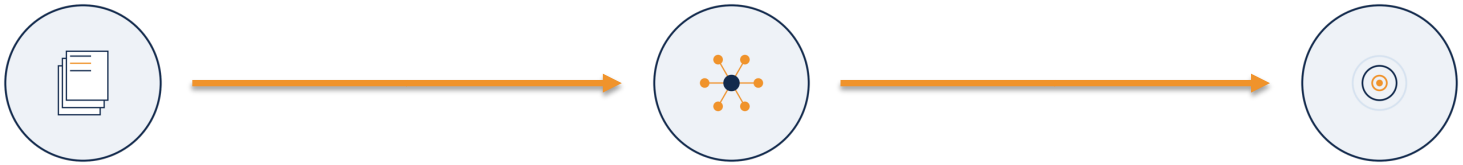
State	Crop	CostCultivation	CostCultivation2	Production	Yield	Temperature	RainFall Annual	Price
Uttar Pradesh	ARHAR	9794.05	9800.25	1941.55	9.83	28.96	3373.2	19589.1
Karnataka	ARHAR	10593.15	10594.15	2172.46	7.47	29.22	3520.7	21187.3
Gujarat	ARHAR	13468.82	13469.82	1898.3	9.59	28.47	2957.4	26938.64
Andhra Pradesh	ARHAR	17051.66	17052.66	3670.54	6.42	28.49	3079.6	34104.32
Maharashtra	ARHAR	17130.55	17131.55	2775.8	8.72	28.3	2566.7	34262.1
Maharashtra	COTTON	23711.44	23712.44	2539.47	12.69	28.73	2534.4	47423.88
Punjab	COTTON	29047.1	29048.1	2003.76	24.39	28.65	3347.9	58095.2
Andhra Pradesh	COTTON	29140.77	29141.77	2509.99	17.83	28.83	3576.4	58282.54
Gujarat	COTTON	29616.09	29617.09	2179.26	19.05	28.38	2899.4	59233.18
Haryana	COTTON	29918.97	29919.97	2127.35	19.9	28.53	2687.2	59838.94
Rajasthan	GRAM	8552.69	8553.69	1691.66	6.83	28.62	2960.5	17106.38
Madhya Pradesh	GRAM	9803.89	9804.89	1551.94	10.29	28.95	2365.8	19608.78
Uttar Pradesh	GRAM	12833.04	12834.04	1882.68	10.93	28.67	2957.8	25667.08
Maharashtra	GRAM	12985.95	12986.95	2277.68	8.05	28.66	2741.3	25972.9
Andhra Pradesh	GRAM	14421.98	14422.98	1559.04	16.69	28.94	2937.5	28844.96
Karnataka		13647.1	13648.1	3484.01	4.71	28.82	2612.4	27295.2



¿Qué puede hacer el Machine Learning?



Diagnóstico médico

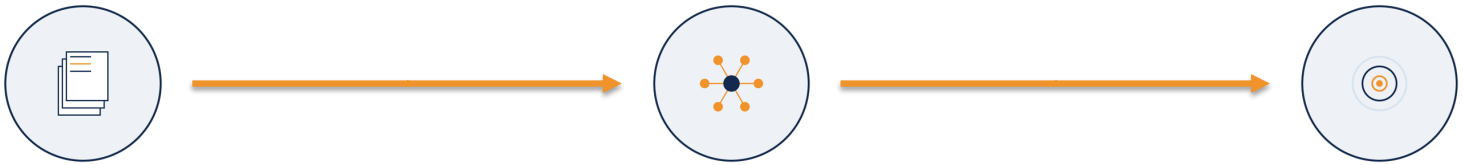


Historial de pacientes con diagnóstico conocido

Modelo entrenado

Predicción: ¿Benigno o maligno?

Éxito académico



Historial de estudiantes con resultado conocido

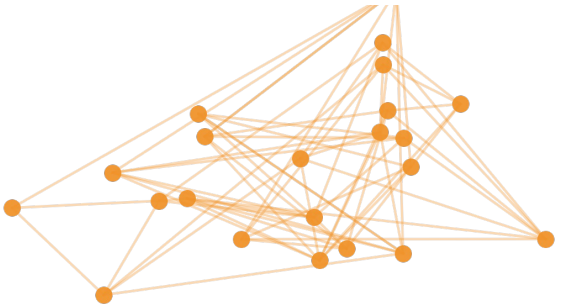
Modelo entrenado

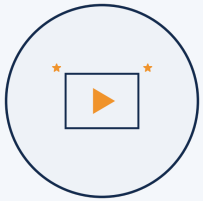
Predicción: ¿Graduate o Dropout?

Ejemplos de ML en tu día a día



Ejemplos de ML en tu día a día

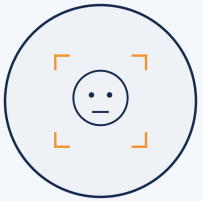




Netflix

Recomendación

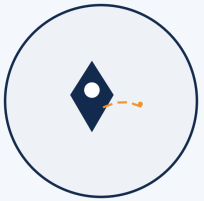
Te sugiere qué ver según lo que ya has visto



Face ID

Clasificación

Decide en milisegundos si eres tú quien desbloquea el móvil



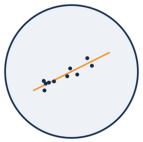
Google Maps

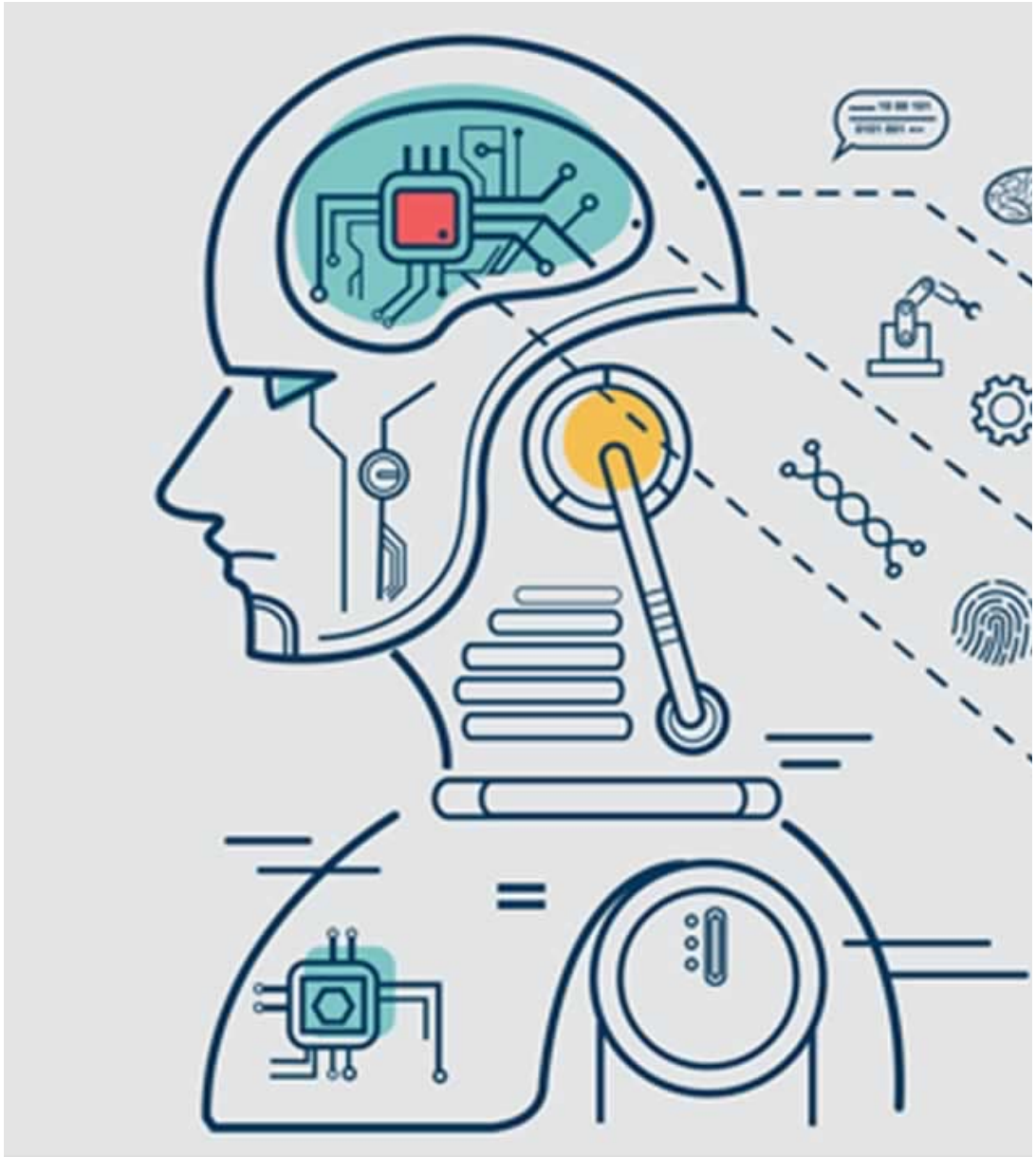
Regresión

Predice cuántos minutos tardarás en llegar







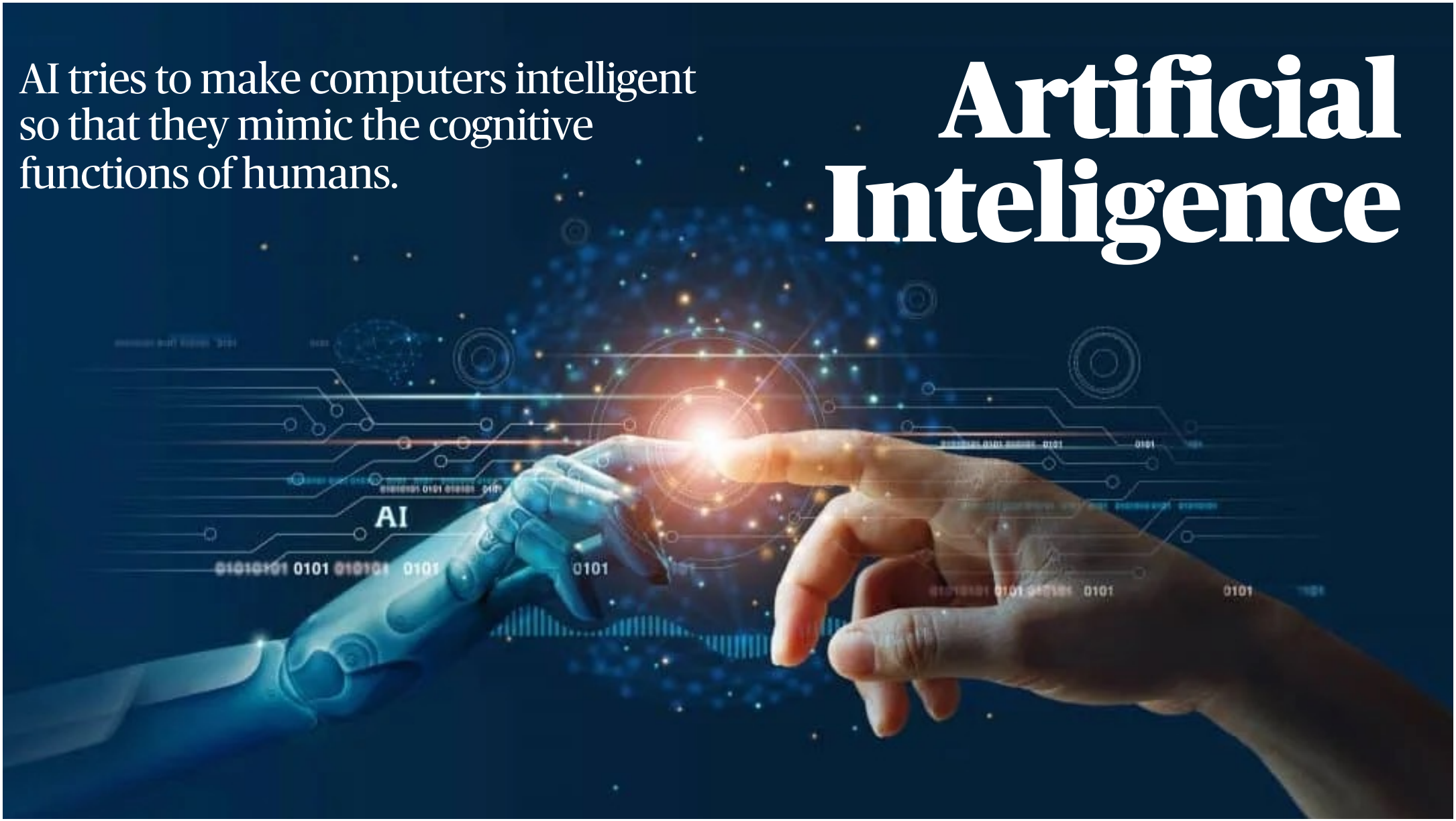


Diferencias entre ...

- Inteligencia Artificial (IA)
- Machine Learning (ML)
- Deep Learning (DL)

AI tries to make computers intelligent so that they mimic the cognitive functions of humans.

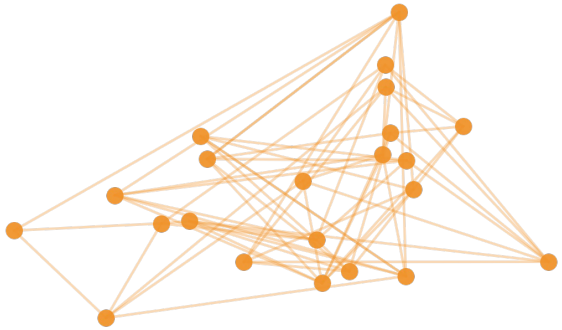
Artificial Inteligence

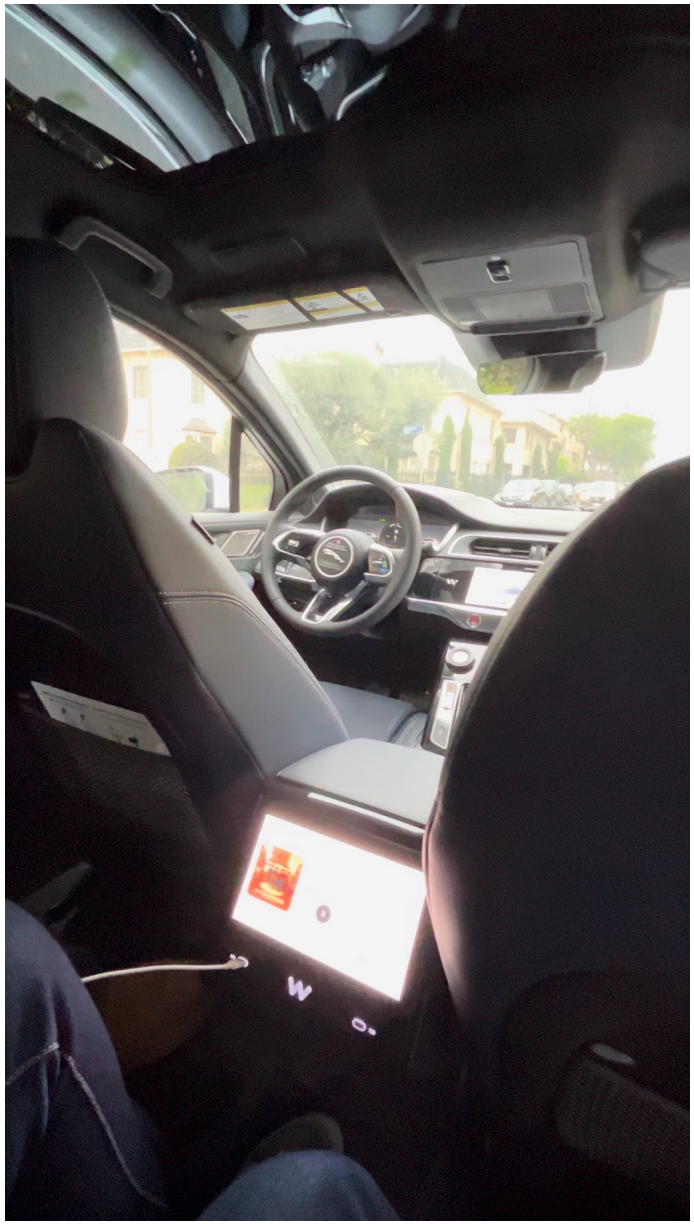


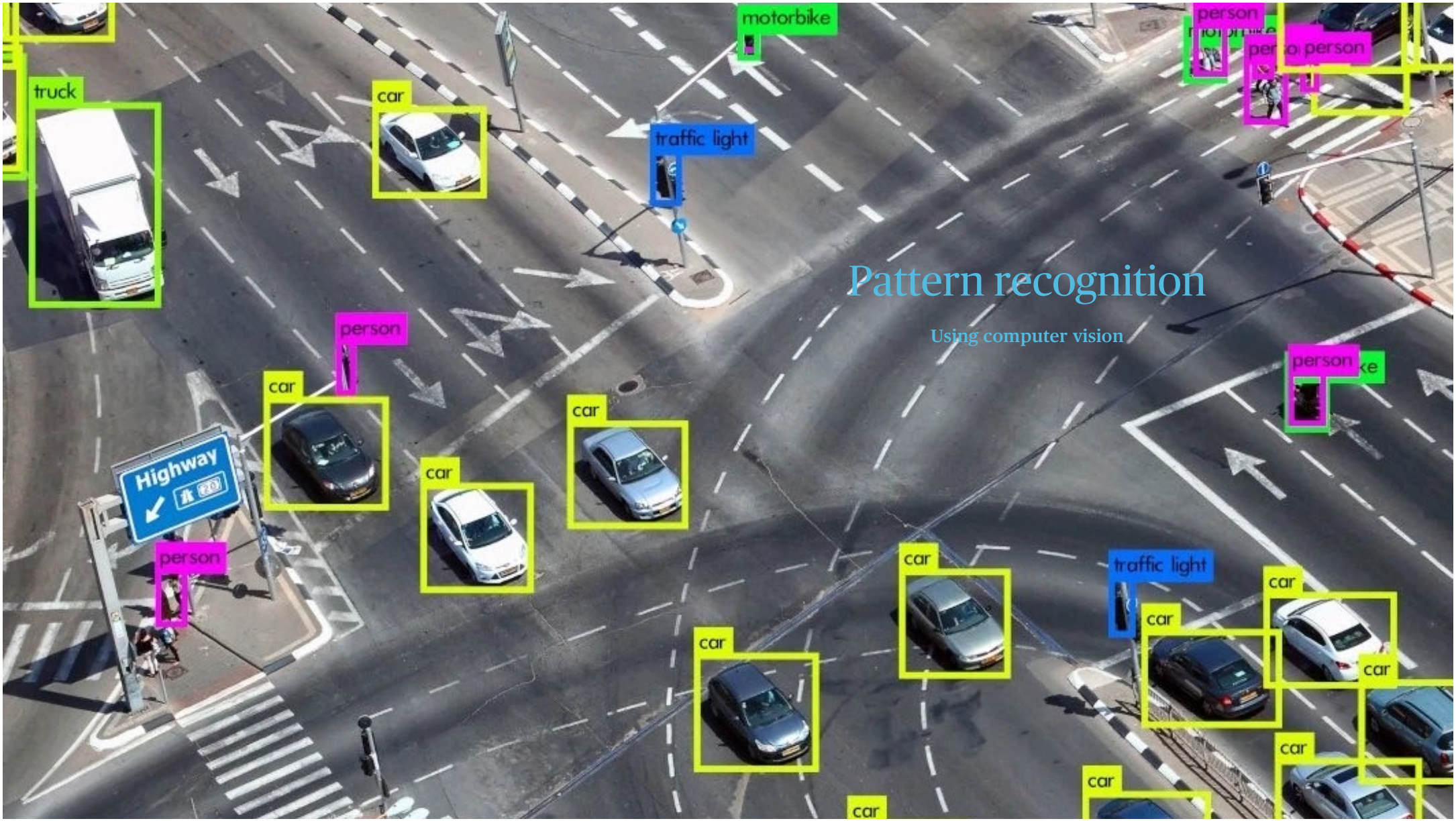

Autopilot in Tesla
Automatic car driving

Autopilot in Waymo

- Automatic car driving in LA



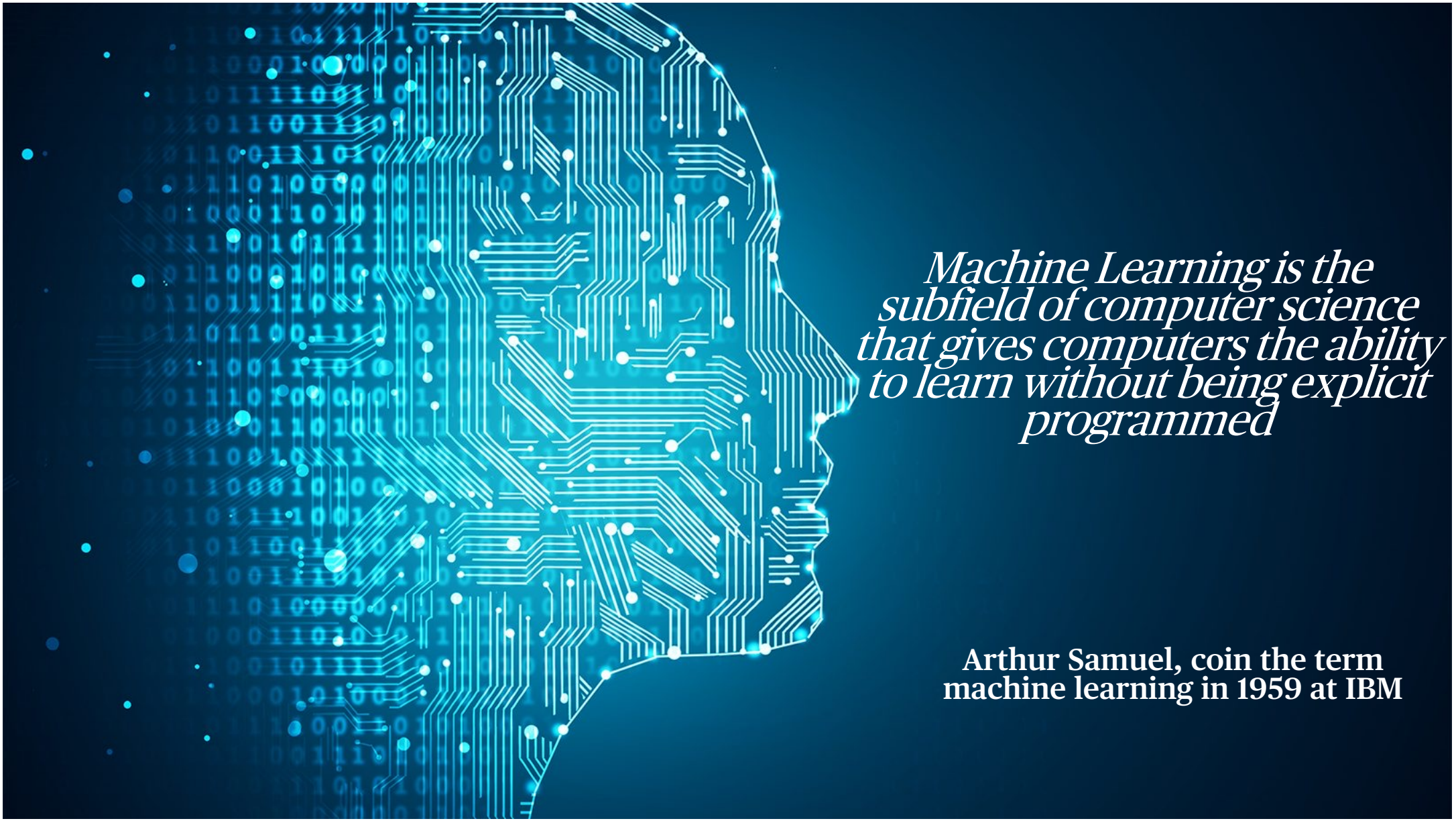


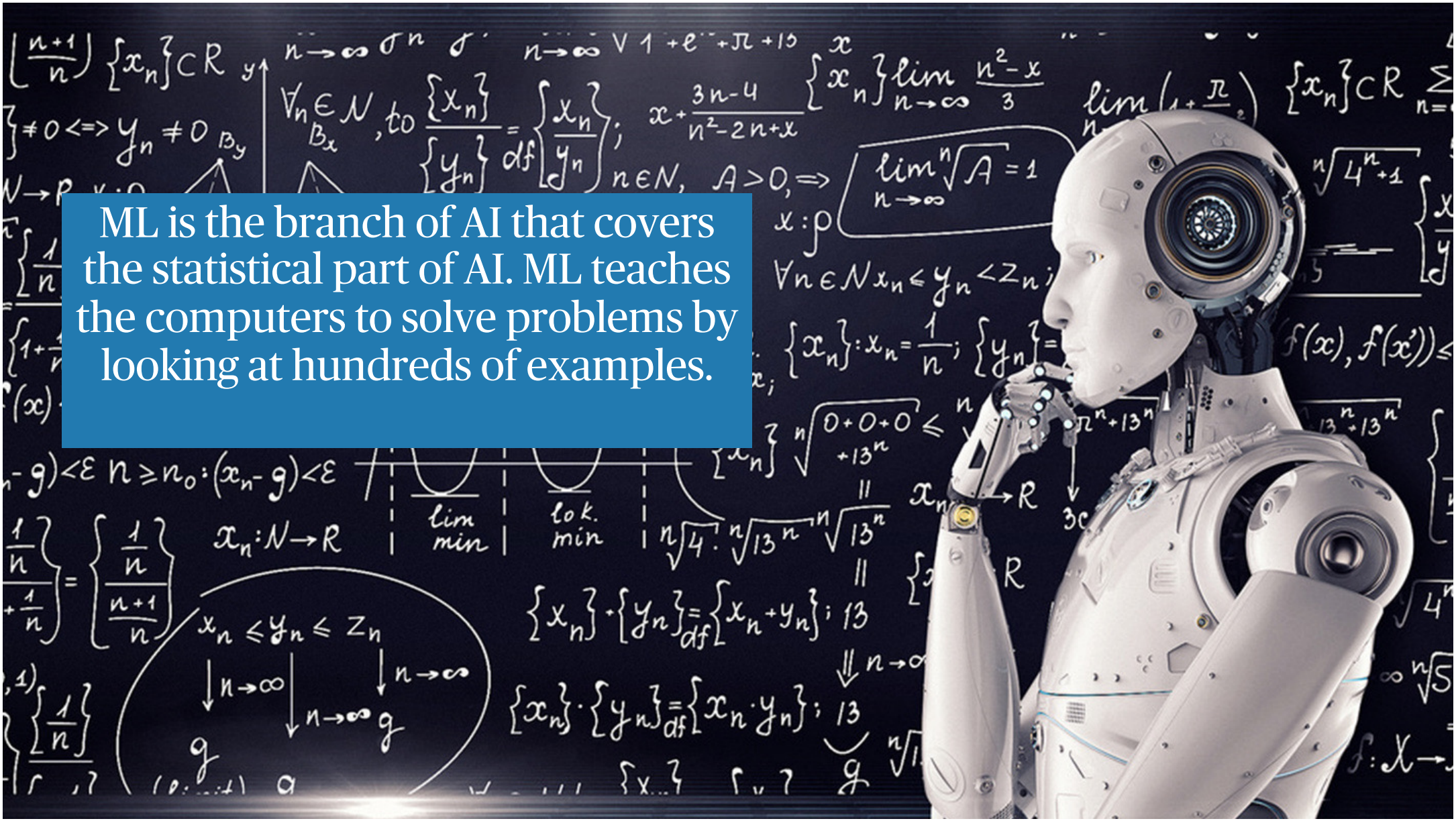

motorbike
person
person
person
truck
car
traffic light
Pattern recognition
Using computer vision
person
car
car
car
Highway
person
person
car
traffic light
car
car
car
car
car
car

Playing games

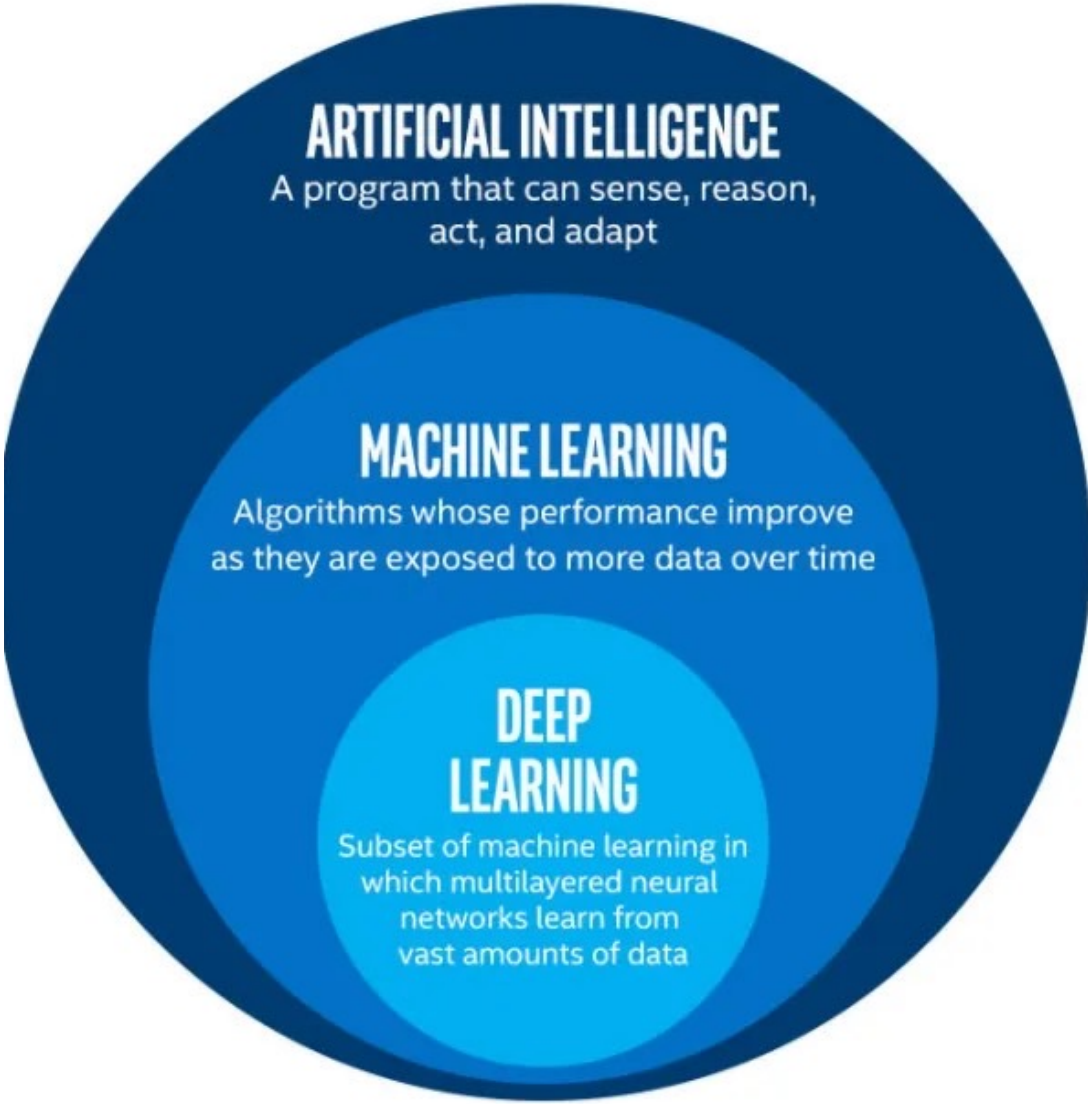
Playing go




Machine Learning is the subfield of computer science that gives computers the ability to learn without being explicit programmed
Arthur Samuel, coin the term machine learning in 1959 at IBM

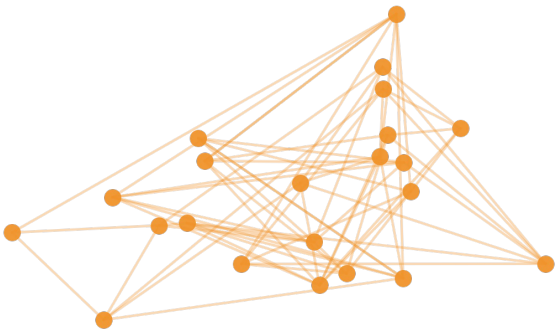

ML is the branch of AI that covers the statistical part of AI. ML teaches the computers to solve problems by looking at hundreds of examples.


11001
DEEP
LEARNING
WWW



Mind map of:

- artificial intelligence,
- machine learning,
- deep learning



Where would ChatGPT fit in?



ChatGPT falls under the umbrella of Deep Learning, and more specifically, it is:

- A Deep Learning model
- Based on deep neural networks
- Of the Transformer type
- Trained as a Large Language Model (LLM)

Classical Machine Learning

Task Driven

Data Driven

Supervised Learning

(Pre Categorized Data)

Classification

(Divide the socks by Color)

Eg. Identity Fraud Detection

Regression

(Divide the Ties by Length)

Eg. Market Forecasting

Unsupervised Learning

(Unlabelled Data)

Clustering

(Divide by Similarity)

Eg. Targeted Marketing

Association

(Identify Sequences)

Eg. Customer Recommendation

Dimensionality Reduction

(Wider Dependencies)

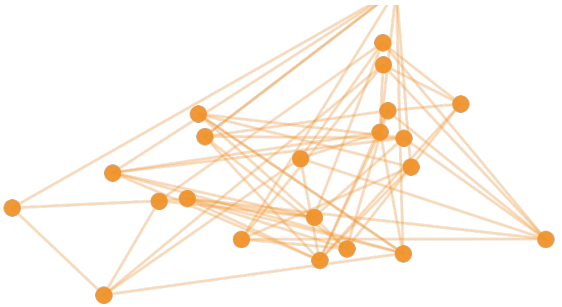
Eg. Big Data Visualization

Obj: Predications & Predictive Models

Pattern/ Structure Recognition

CÓMO APRENDE UN MODELO (I)

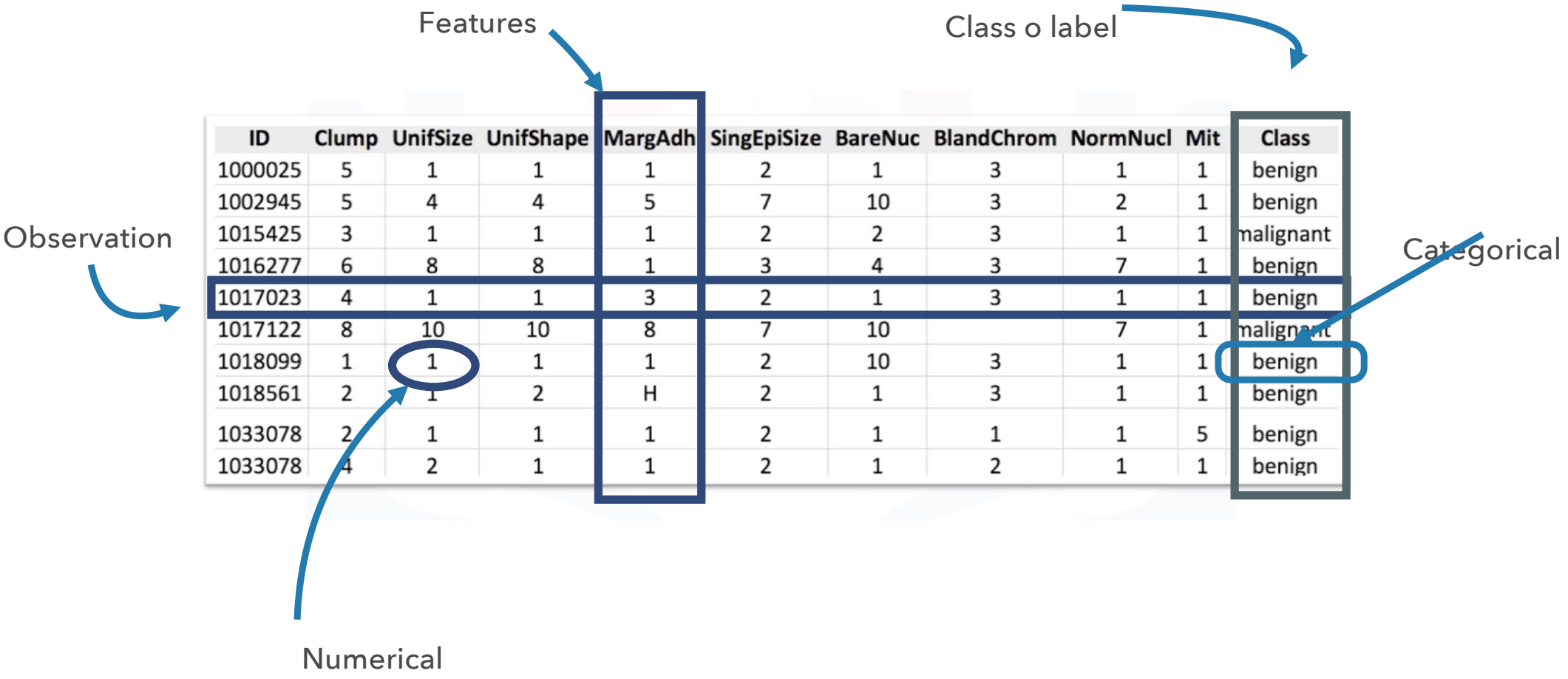
Aprendizaje supervisado





En los modelos supervisados, entrenamos al modelo mostrándole ejemplos de un dataset ya etiquetado: para cada caso, ya sabemos cuál es la respuesta correcta.

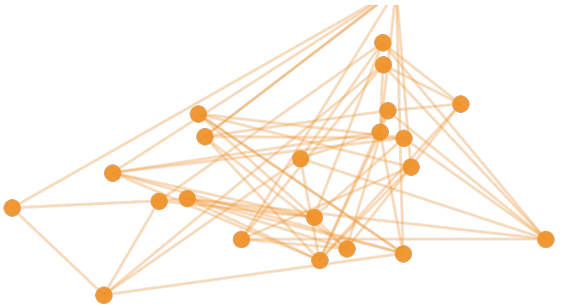
¿Cómo es un dataset etiquetado?



ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

CÓMO APRENDE UN MODELO (I)

Te voy a enseñar algo...



1 → 2

2 → 4

5 → 10

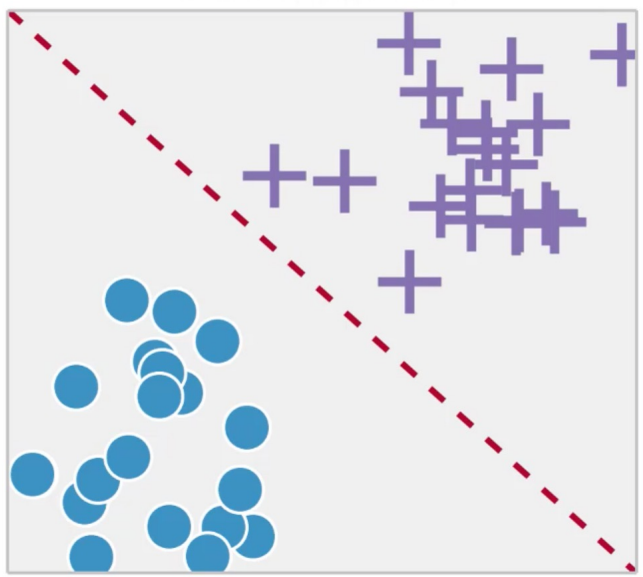
6 → 12

10 → ?

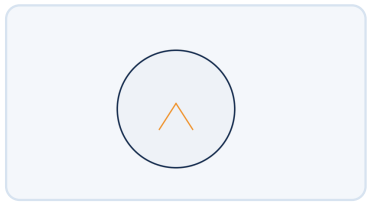
¿Qué número dirías? Acabas de aprender un patrón a partir de ejemplos: eso es, en esencia, aprendizaje supervisado.

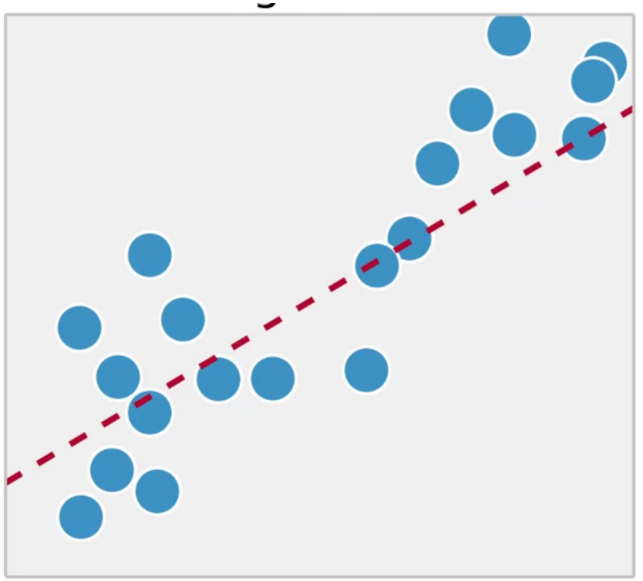
CÓMO APRENDE UN MODELO (I)

Dentro de lo supervisado: clasificación o regresión



Clasificación



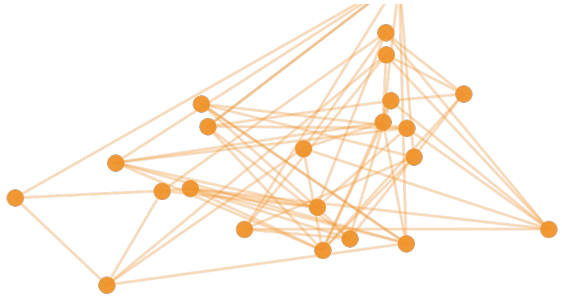


Regresión



CÓMO APRENDE UN MODELO (II)

Aprendizaje no supervisado





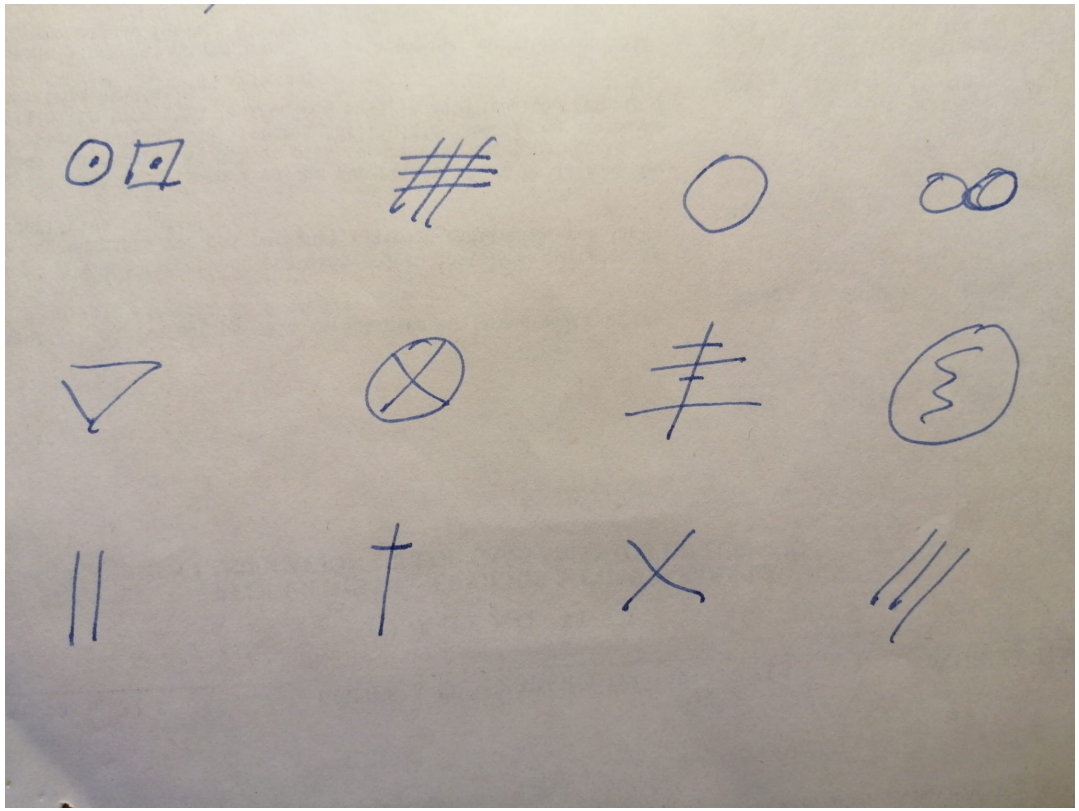
Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
1	41	2	6	19	0.124	1.073	NBA001	6.3
2	47	1	26	100	4.582	8.218	NBA021	12.8
3	33	2	10	57	6.111	5.802	NBA013	20.9
4	29	2	4	19	0.681	0.516	NBA009	6.3
5	47	1	31	253	9.308	8.908	NBA008	7.2
6	40	1	23	81	0.998	7.831	NBA016	10.9
7	38	2	4	56	0.442	0.454	NBA013	1.6
8	42	3	0	64	0.279	3.945	NBA009	6.6
9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6

Aquí no le damos ninguna etiqueta al modelo: lo dejamos trabajar solo, para que descubra por sí misma una estructura que a simple vista no habríamos visto.

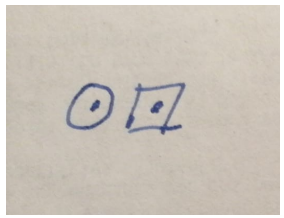
CÓMO APRENDE UN MODELO (II)

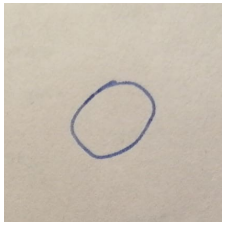
Un dataset sin etiquetar: ¿cuántos grupos ves?

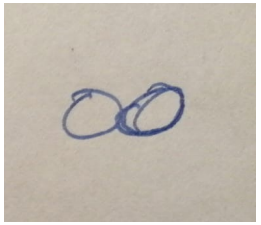
Imagina que eres arqueólogo y encuentras símbolos de una lengua desconocida.
No sabes nada de ellos, ¿pero podrías agruparlos?

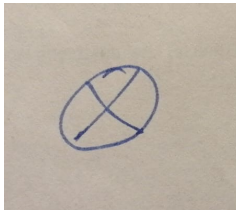


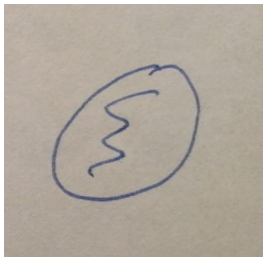
Polygonal language?

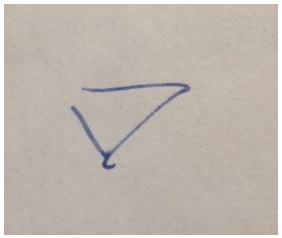






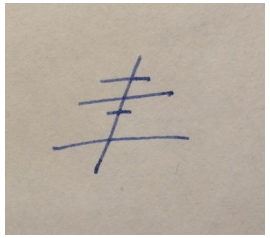


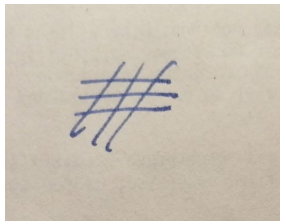


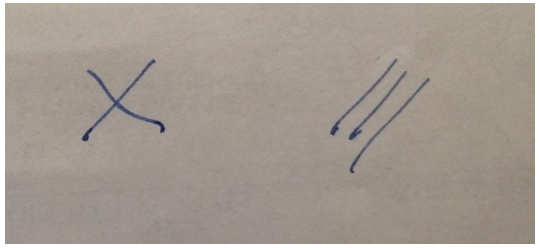


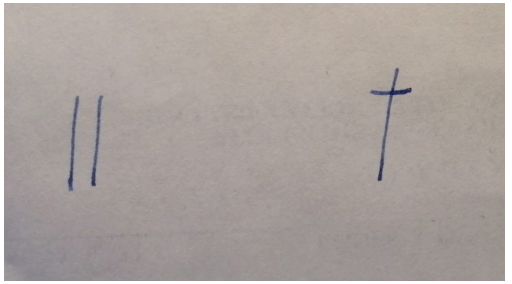
Two clusters

Linear language?

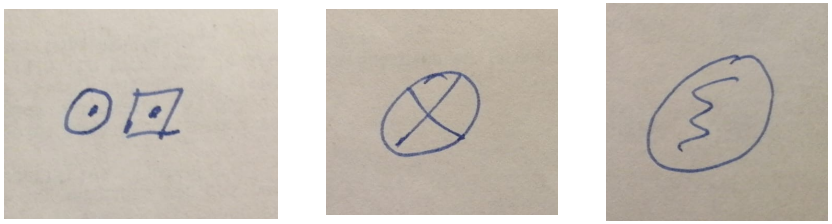




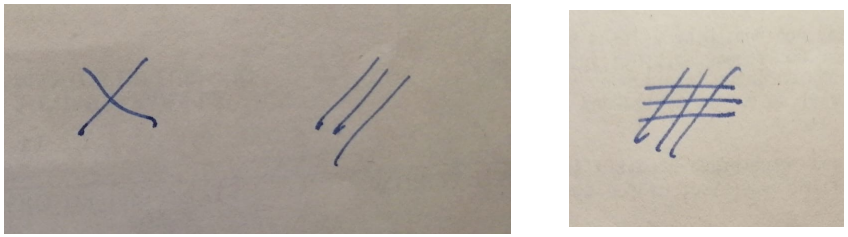




Fulled polygonal language?

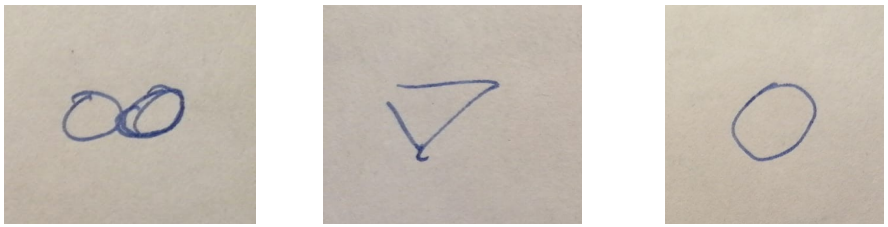


Diagonal Linear language?

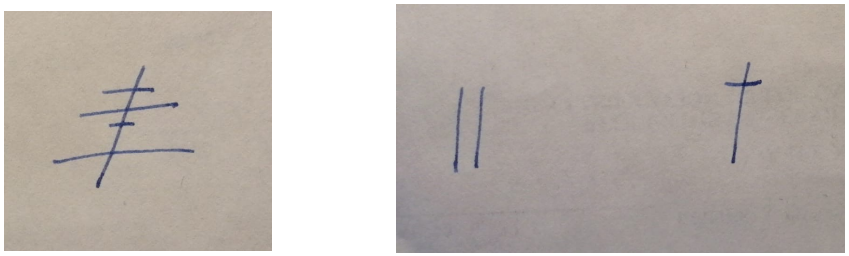


Four clusters?

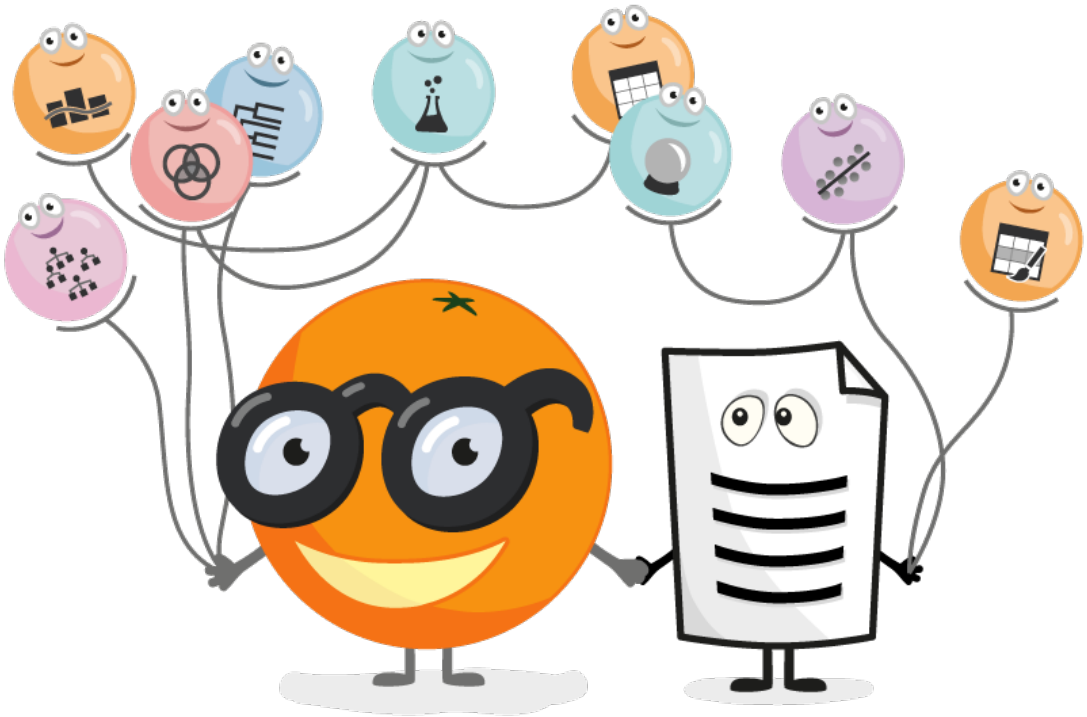
Empty polygonal language?



Straight Linear language?



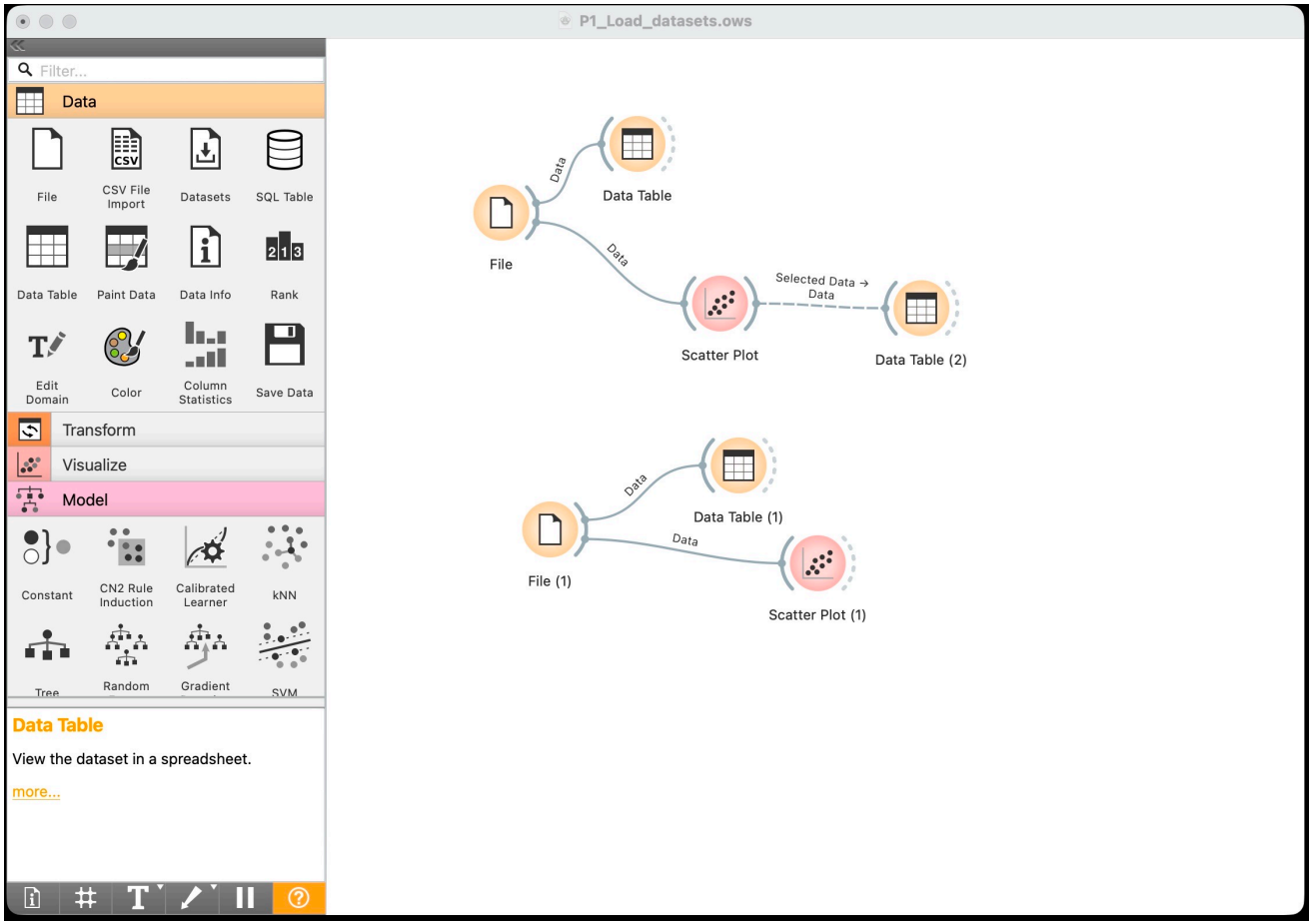
Primer programa con Orange 3

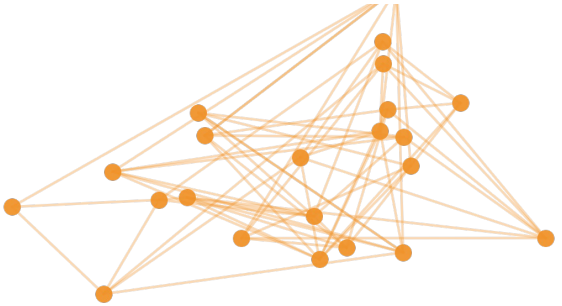


Tu turno. Primeros pasos

- Importamos File ->
- Data Table
- Scatter Plot
- Selected data table

Primer programa con Orange 3





Tu turno. Primeros pasos

- Importamos File ->
- Data Table
- Scatter Plot
- Selected data table

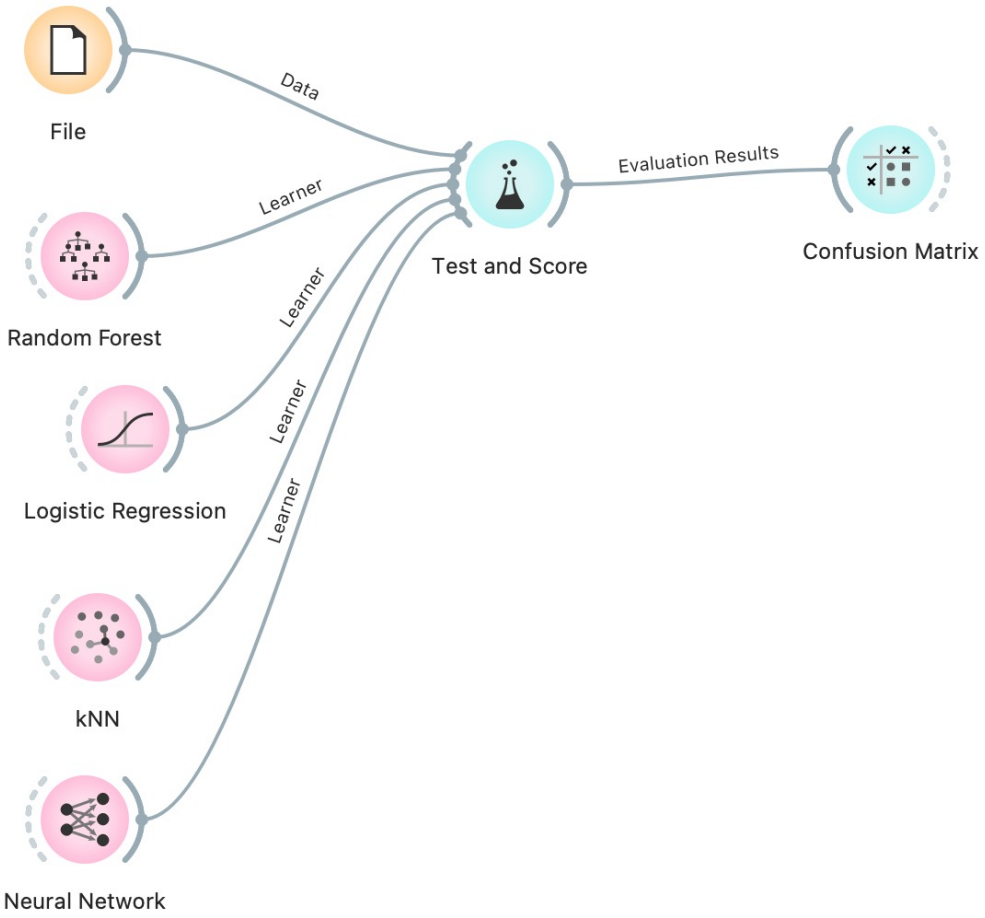
Classification problem in ML

ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

Classification

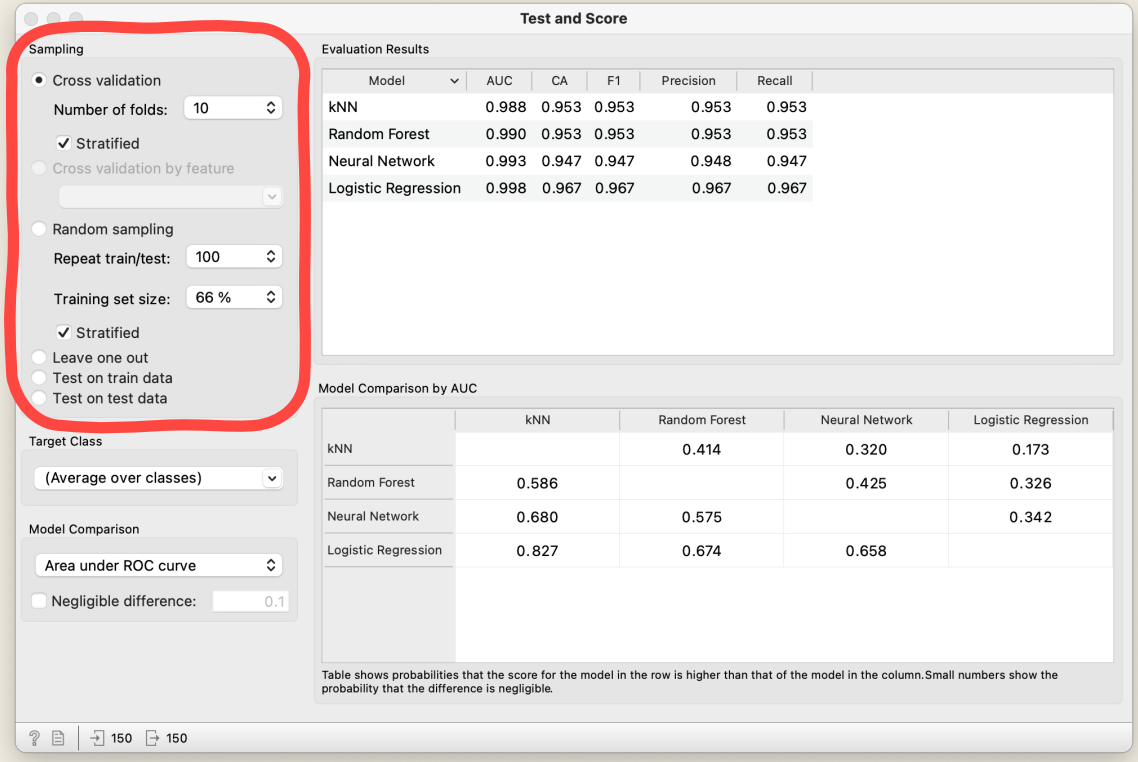
In classification, the goal is to predict a *class label*, which is a choice from a predefined list of possibilities.

- Binary Classification
- Multiclass Classification



Scoring

- K-cross validation
- Test & Score
- Confusion Matrix



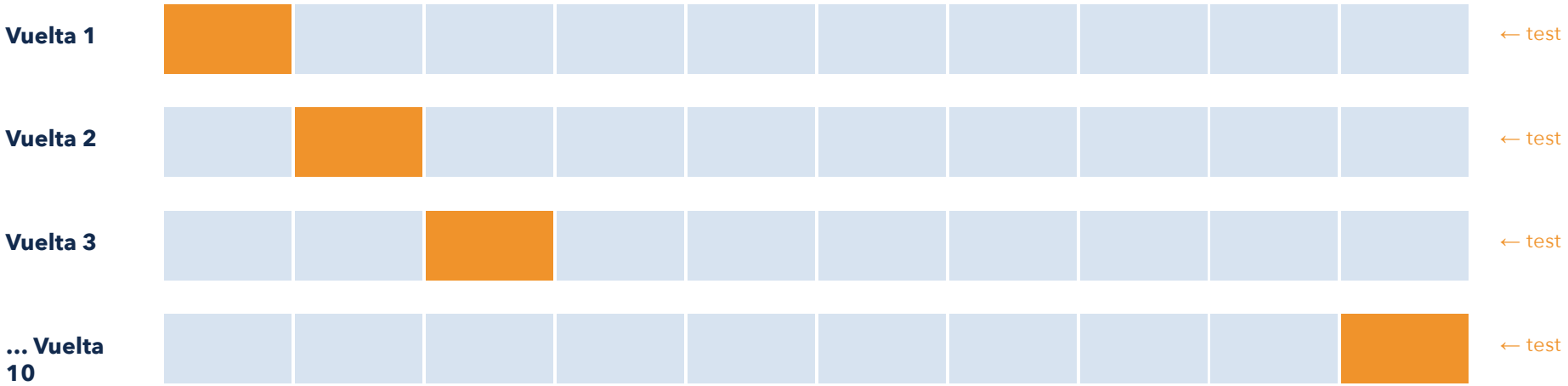
Sampling

- Cross validation
- Random Samplng
- Other options

SAMPLING: ¿CÓMO DIVIDIMOS LOS DATOS?

Cross-validation: validación cruzada en k folds

Cada color representa un "fold": naranja = test de esa vuelta, azul claro = entrenamiento

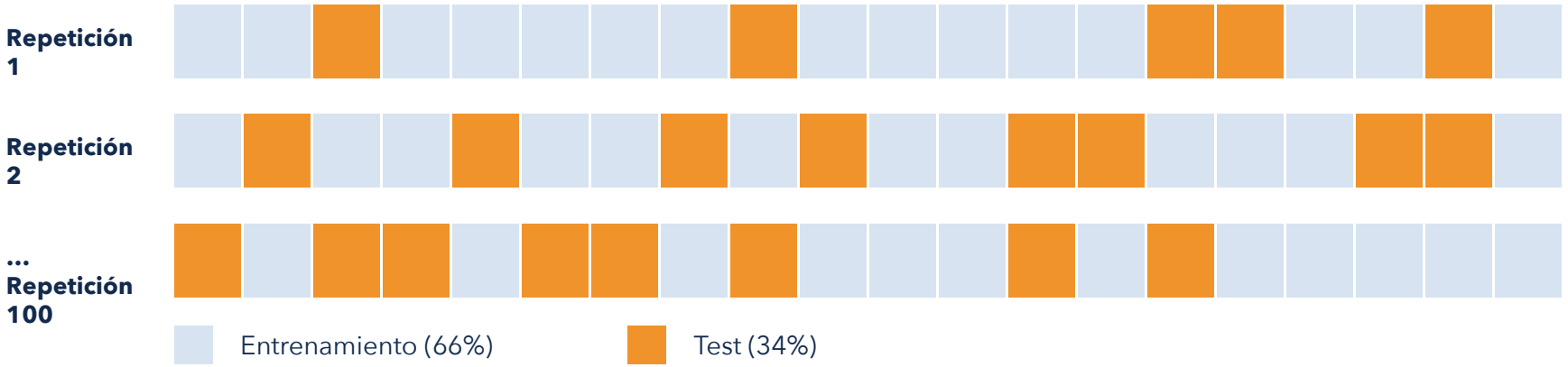


Resultado final = media de las 10 puntuaciones

SAMPLING: ¿CÓMO DIVIDIMOS LOS DATOS?

Random sampling: repeticiones con reparto aleatorio

En cada repetición, se reparte al azar qué filas van a entrenamiento (66%) y cuáles a test (34%)

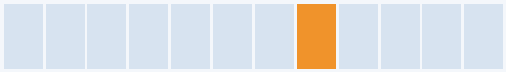


Resultado final = media de las 100 repeticiones

SAMPLING: ¿CÓMO DIVIDIMOS LOS DATOS?

Las otras tres opciones de la columna

Leave one out



k = número de filas (1 fila de test cada vez)

El caso extremo de cross-validation: se deja fuera 1 sola fila cada vez. Exhaustivo, pero caro con muchos datos.

Test on train data

TODOS LOS DATOS

entrena →

evalúa →

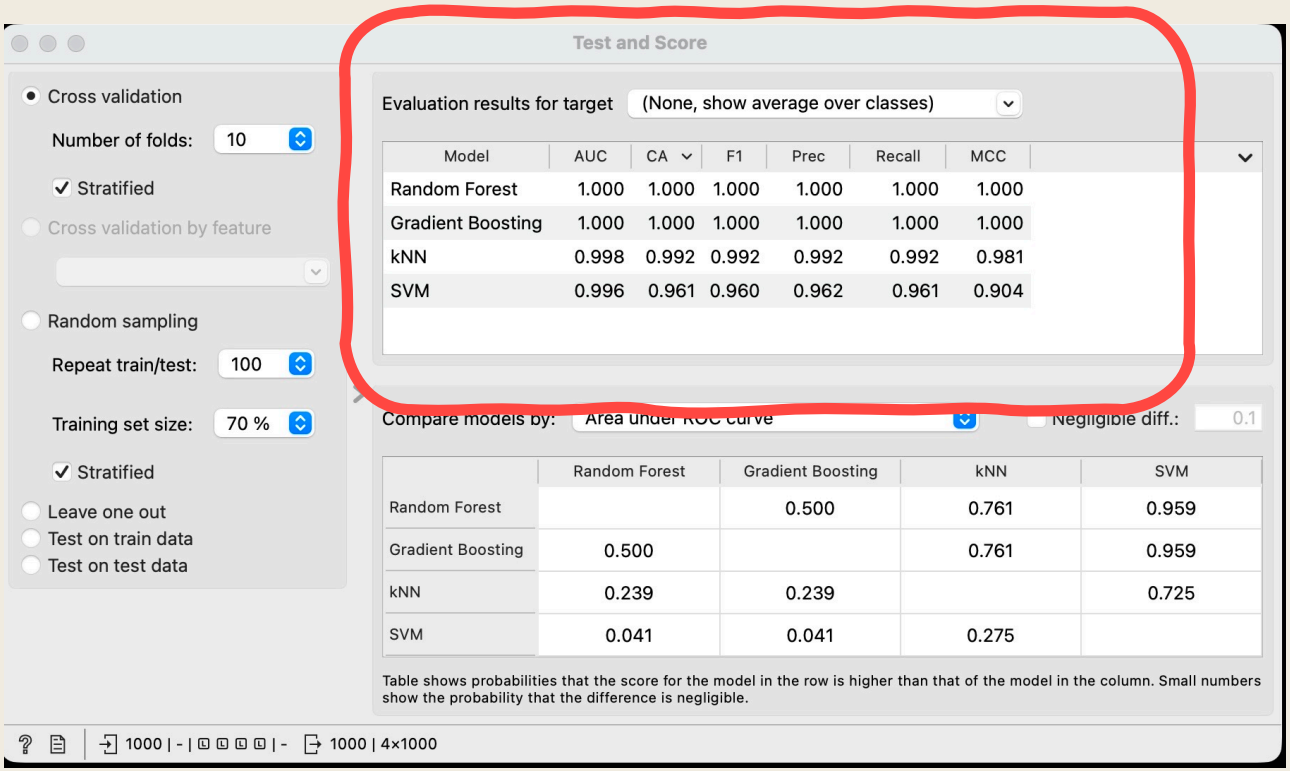
mismos datos

Entrena y evalúa con los mismos datos. Da resultados demasiado optimistas – útil solo para ver el efecto del sobreajuste.

Test on test data

train.tab + **test.tab**

Para cuando ya tienes dos ficheros separados de antemano: uno de entrenamiento y otro de test real.



Evaluation metrics in classification

- We are talking about some metrics:
- Confusion matrix

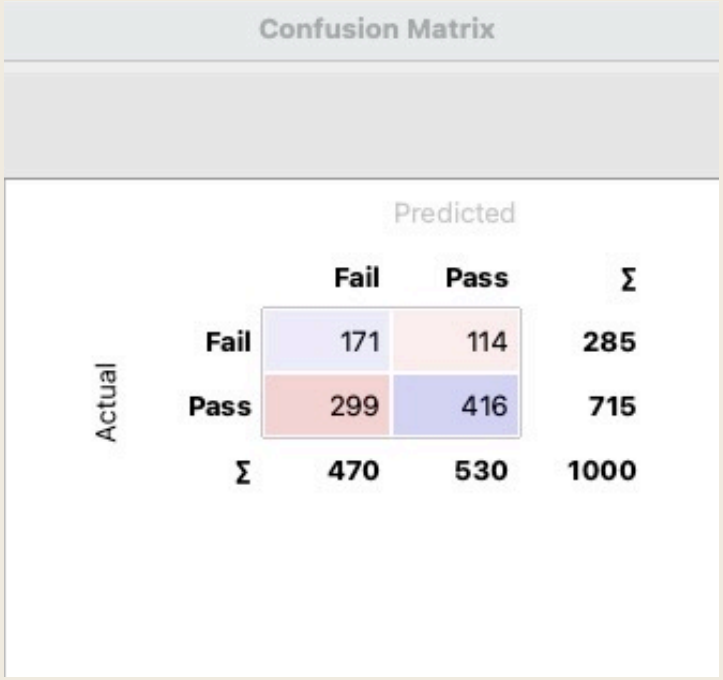
Confusion Matrix

TP = True Positives
FN = False Negatives
FP = False Positives
TN = True Negatives

$$\textbf{\textit{Precision}} = \frac{TP}{TP+FP}$$

$$\textbf{\textit{Recall}} = \frac{TP}{TP+FN}$$

$$\textbf{\textit{F1-score}}\ [0,1] = \frac{2*Precision*Recall}{Precision+Recall}$$



Classification Accuracy is the proportion of correctly classified examples

$$CA = \frac{TP + TN}{Total}$$

$$MCC = \frac{TP\ x\ TN - FP\ x\ FN}{sqrt((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$$

CÓMO SE MIDE UN MODELO

Los scores salen directamente de esos 4 números

CA

(TP+TN) / total

(416+171)/1000 = 0.587

Precision

TP / (TP+FP)

416/(416+114) = 0.785

Recall

TP / (TP+FN)

416/(416+299) = 0.582

F1

media armónica de Precision y Recall

= 0.668

MCC

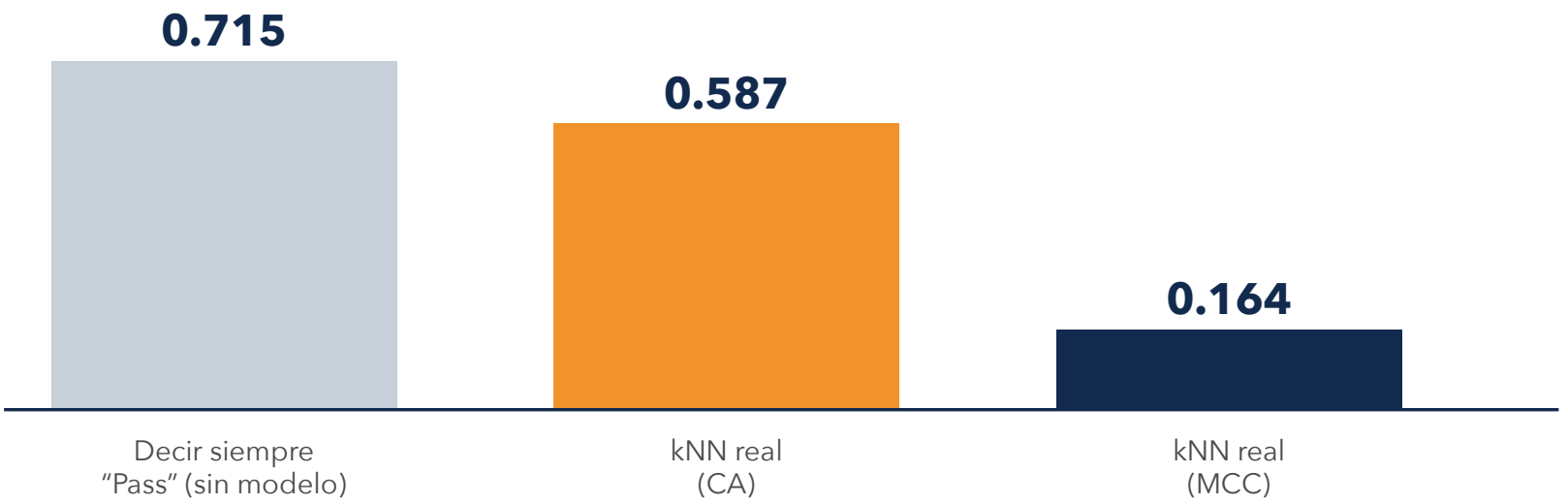
$$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

= 0.164

CÓMO SE MIDE UN MODELO

¿Por qué no basta con mirar el Accuracy?

El 71.5% de los estudiantes aprueba. Un modelo que siempre dijera “Pass” acertaría el 71.5% sin aprender nada – y este kNN se queda en 58.7%: ¡peor que no hacer nada! El MCC (0.164) confirma que apenas hay señal real en las variables demográficas.



CÓMO SE MIDE UN MODELO

El MCC pregunta: ¿Tus predicciones son verdad, o podrían salir por puro azar?

El modelo predijo "Pass" para 530 de los 1000 estudiantes.

Si esas 530 predicciones fueran completamente aleatorias . ¿cuántos aciertos (TP) esperarías solo por chance?

Como el 71.5% de todo el mundo aprueba, de esas 530 personas "elegidas al azar" un 71.5% aprobaría de todas formas:

530 × 715/1000 ≈ 379 aciertos esperados solo por azar

Nuestro modelo real consigue 416 aciertos. Es decir, el modelo solo mejora el resultado del puro azar en 37 personas de 1000 — una diferencia muy pequeña.

Un MCC de 0.164 te está diciendo justo eso: "apenas por encima de adivinar sin ninguna información real".



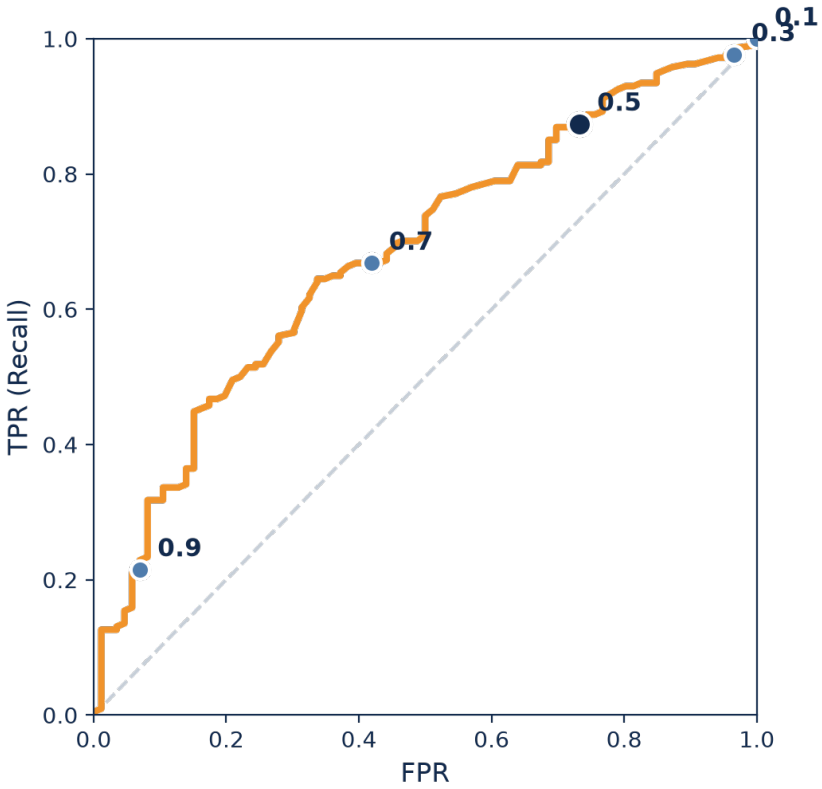
CÓMO SE MIDE UN MODELO

La curva ROC: moviendo el umbral de decisión

Cada umbral de decisión da una matriz de confusión distinta — y por tanto un punto distinto de TPR/FPR.

Umbral	TPR (Recall)	FPR
0.90	0.215	0.070
0.70	0.668	0.419
0.50	**0.874**	**0.733**
0.30	0.977	0.965
0.10	1.000	1.000

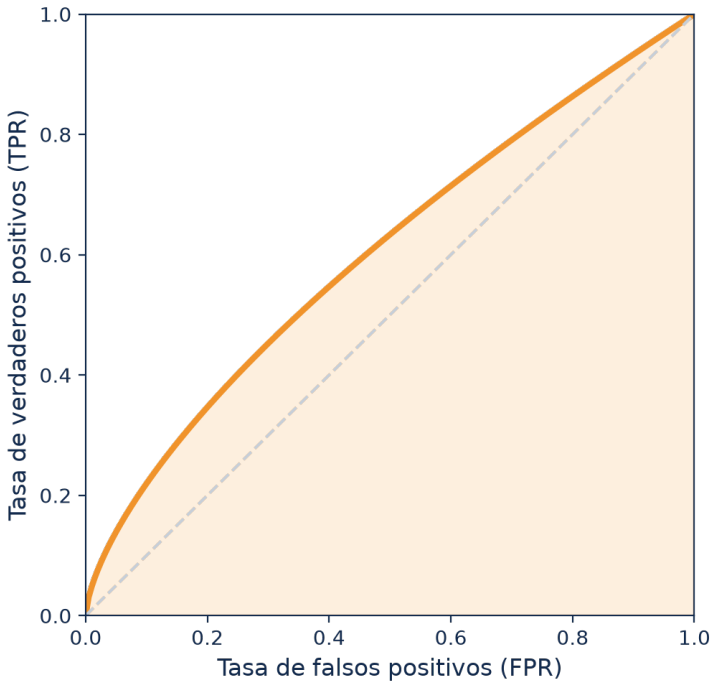
Fila naranja = el umbral por defecto (0.5), la misma matriz de confusión que ya vimos.



CÓMO SE MIDE UN MODELO

AUC: ¿y si miro todos los umbrales a la vez?

La matriz de confusión depende del umbral de decisión que uses. La curva ROC dibuja, para cada umbral posible, la tasa de aciertos frente a la de falsas alarmas. El AUC es, simplemente, el área bajo esa curva — aquí, el AUC real del kNN según Orange3 (widget Test and Score).



AUC = 0.603

0.5 = adivinar al azar

1.0 = clasificador perfecto

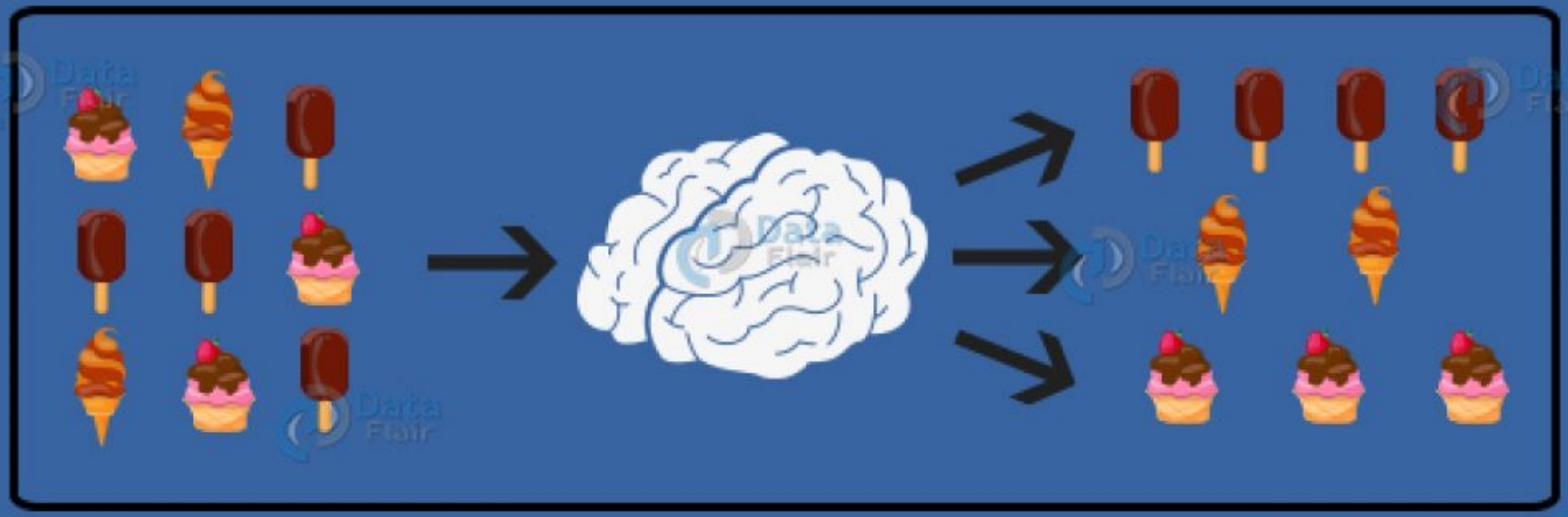
Classification Algorithms

Machine Learning Classification Algorithms

Logistic Regression

Naive Bayes

Decision Tree



Support Vector Machines

Random Forest

K-Nearest Neighbou

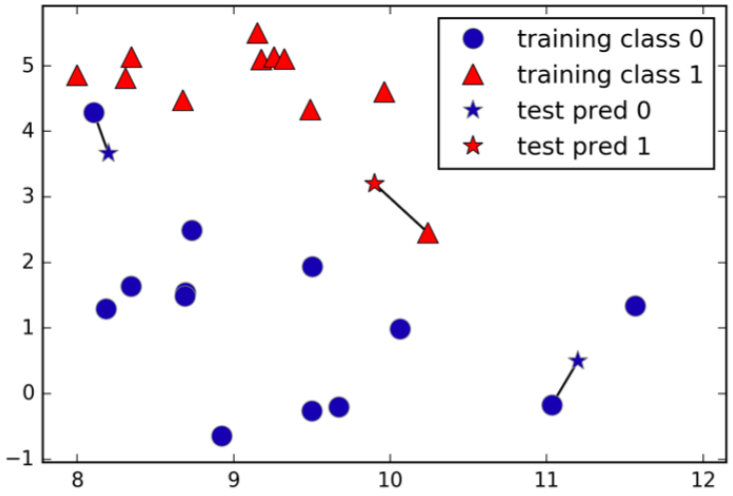


Figure 2-4. Predictions made by the one-nearest-neighbor model on the forge dataset

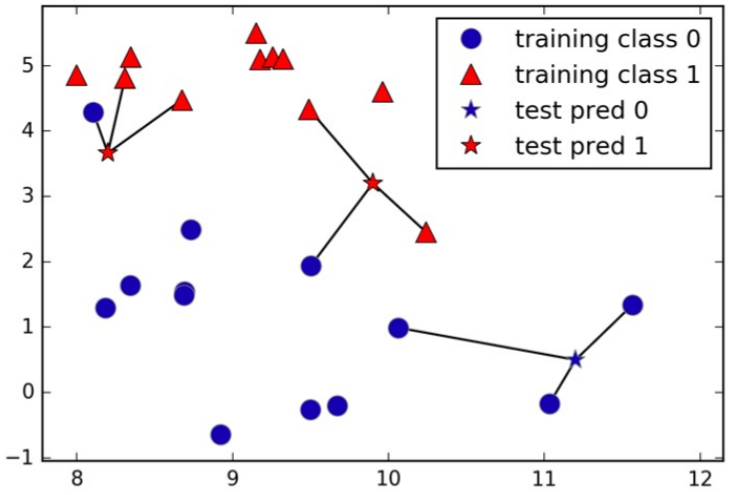
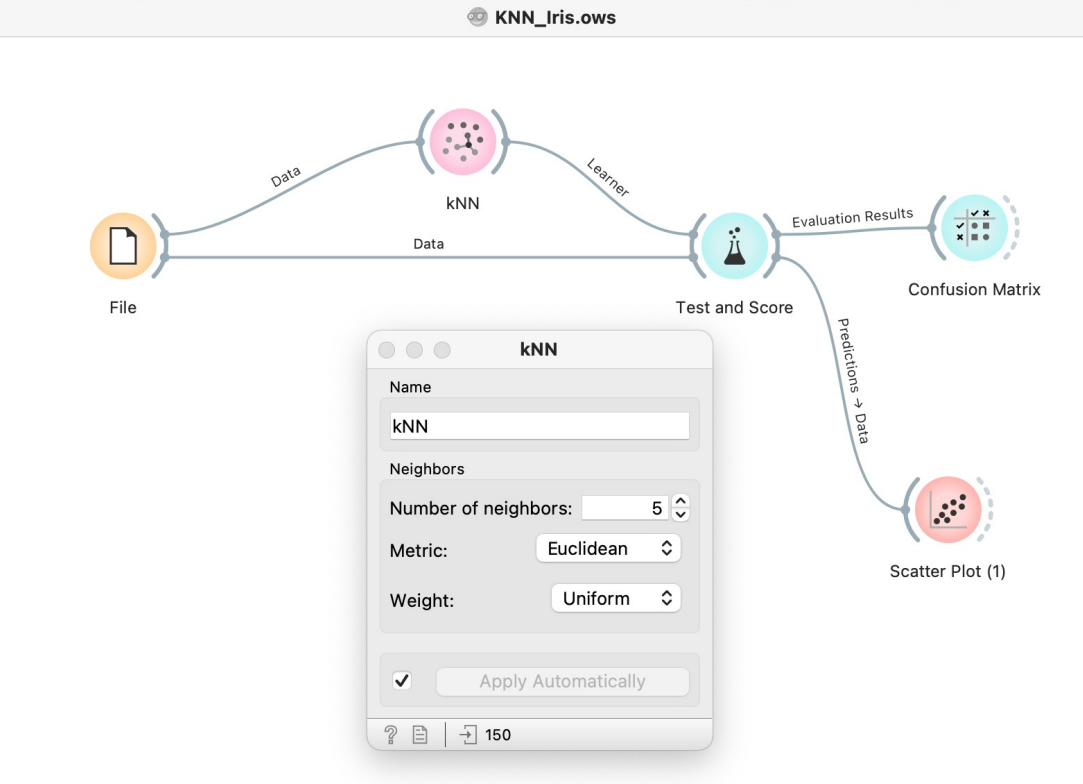


Figure 2-5. Predictions made by the three-nearest-neighbors model on the forge dataset

K-Nearest Neighbors

- Building the k-NN algorithm consists only of storing the training dataset.
- To make a prediction for a new data point, the algorithm finds the closest data points in the training dataset—its "nearest neighbors."



K-nearest Neighbors

- Dropouts
- KNN –> Test Score –> Confusion Matrix
- Different values of K and distances.
- Scatter Plot (KNN) + Box plot

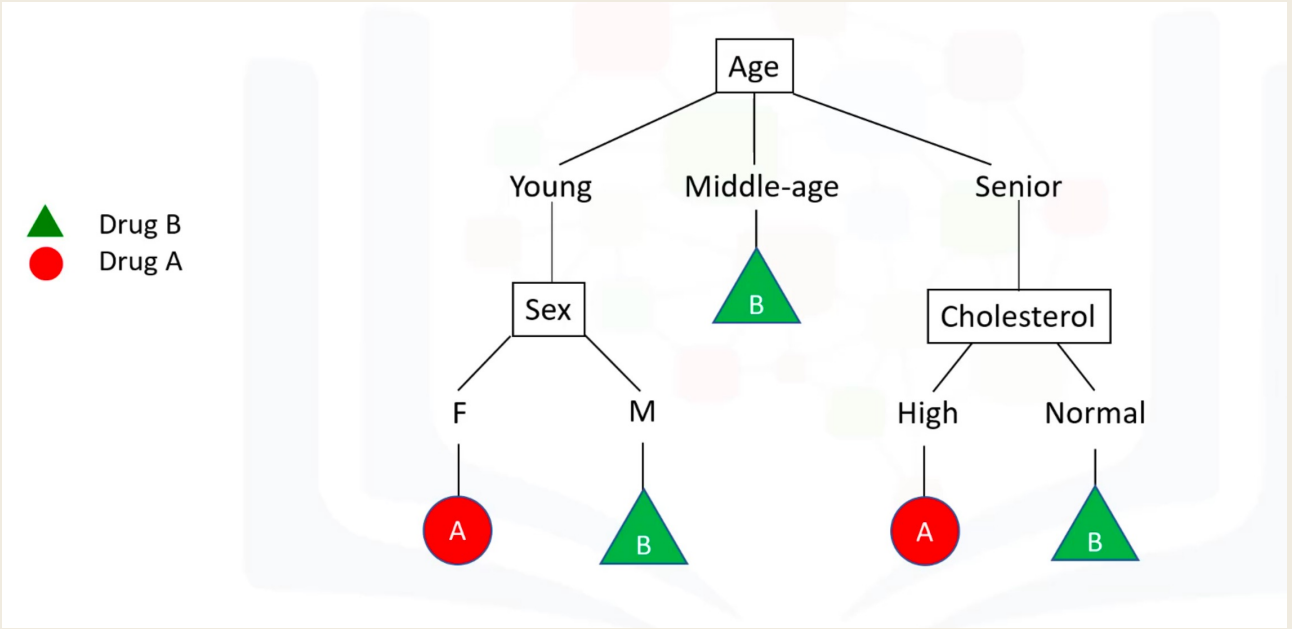
Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	Hiigh	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A
p15	Middle-age	F	Low	Normal	?

Decision Tree

- What exactly is a decision tree?

How to build a decision tree?

1. Choose an attribute from your dataset.
2. Calculate the significance of attribute in splitting of data.
3. Split data based on the value of the best attribute.
4. Go to step 1.



How to calculate the significance?

Information gain is the information that can increase the level of certainty after splitting.

Information Gain = (Entropy before split) – (weighted entropy after split)

In every step, we want to dismiss the impurity of the nodes. Impurity is calculated by Entropy of data in the node

$$E = -\sum_i p_i \log_2(p_i)$$

Entropy is the amount of information disorder. If the samples are completely homogeneous the entropy is zero.

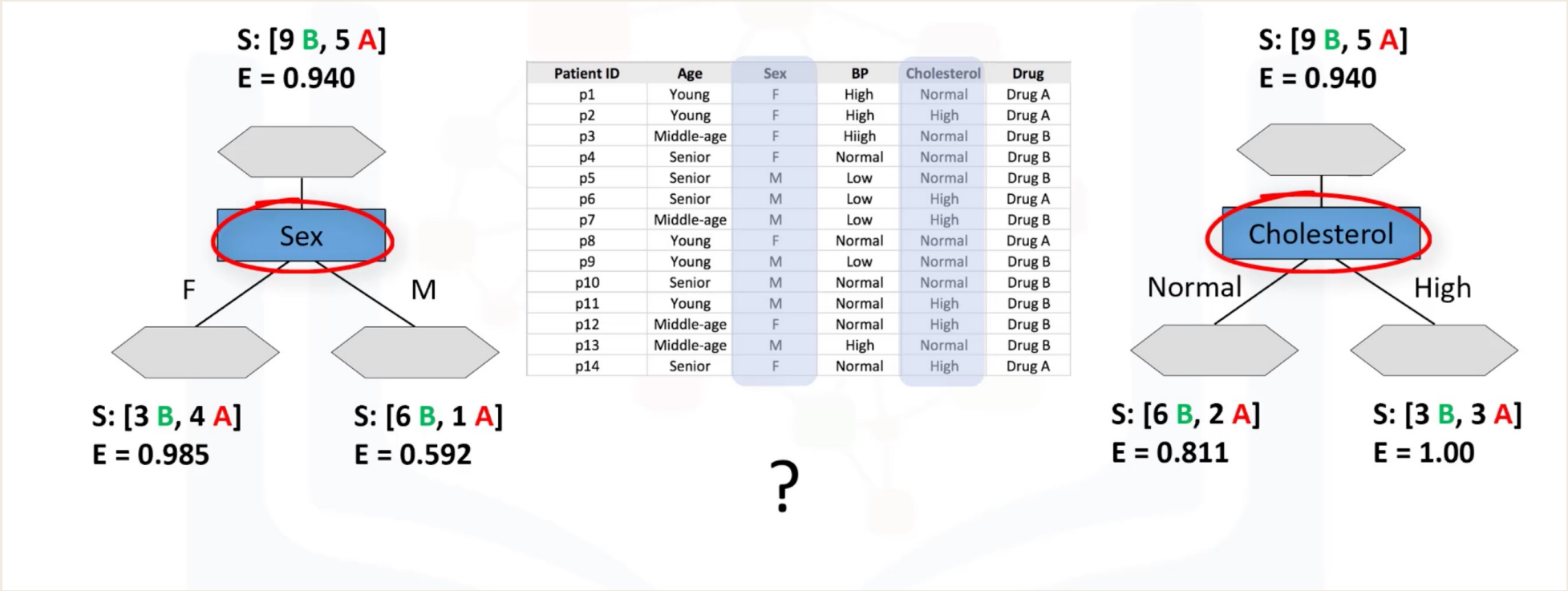
How to calculate the significance?

To compute Gini impurity for a set of items with J clases, suppose $i \in \{1,2,\ldots,J\}$, and let be the fraction of items labeled with class in the set.

$$\mathrm{I}_G(p) = \sum_{i=1}^{J}\left(p_i \sum_{k\neq i} p_k\right) = \sum_{i=1}^{J} p_i(1-p_i) = \sum_{i=1}^{J}\left(p_i - p_i^2\right) = \sum_{i=1}^{J} p_i - \sum_{i=1}^{J} p_i^2 = 1 - \sum_{i=1}^{J} p_i^2$$

This is the other way to measure the significance

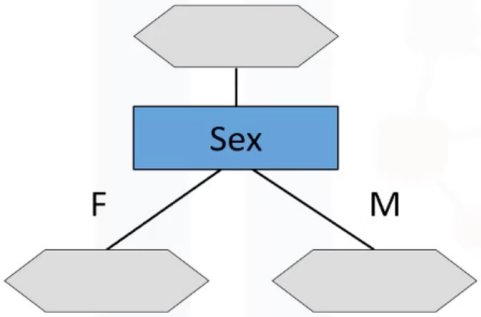
Which attributes is the best?



Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	Hiigh	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A

$$E = -\sum_i p_i \log_2(p_i)$$

S: [9 B, 5 A]
E = 0.940



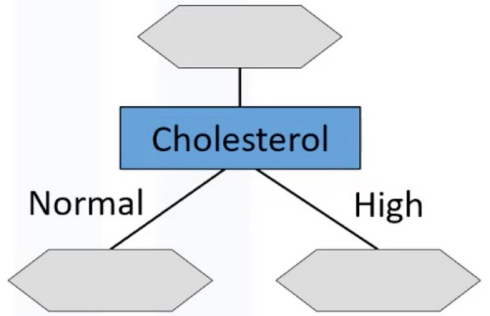
S: [3 B, 4 A]
E = 0.985

S: [6 B, 1 A]
E = 0.592

Patient ID	Age	Sex	BP	Cholesterol	Drug
p1	Young	F	High	Normal	Drug A
p2	Young	F	High	High	Drug A
p3	Middle-age	F	Hiigh	Normal	Drug B
p4	Senior	F	Normal	Normal	Drug B
p5	Senior	M	Low	Normal	Drug B
p6	Senior	M	Low	High	Drug A
p7	Middle-age	M	Low	High	Drug B
p8	Young	F	Normal	Normal	Drug A
p9	Young	M	Low	Normal	Drug B
p10	Senior	M	Normal	Normal	Drug B
p11	Young	M	Normal	High	Drug B
p12	Middle-age	F	Normal	High	Drug B
p13	Middle-age	M	High	Normal	Drug B
p14	Senior	F	Normal	High	Drug A

?

S: [9 B, 5 A]
E = 0.940



S: [6 B, 2 A]
E = 0.811

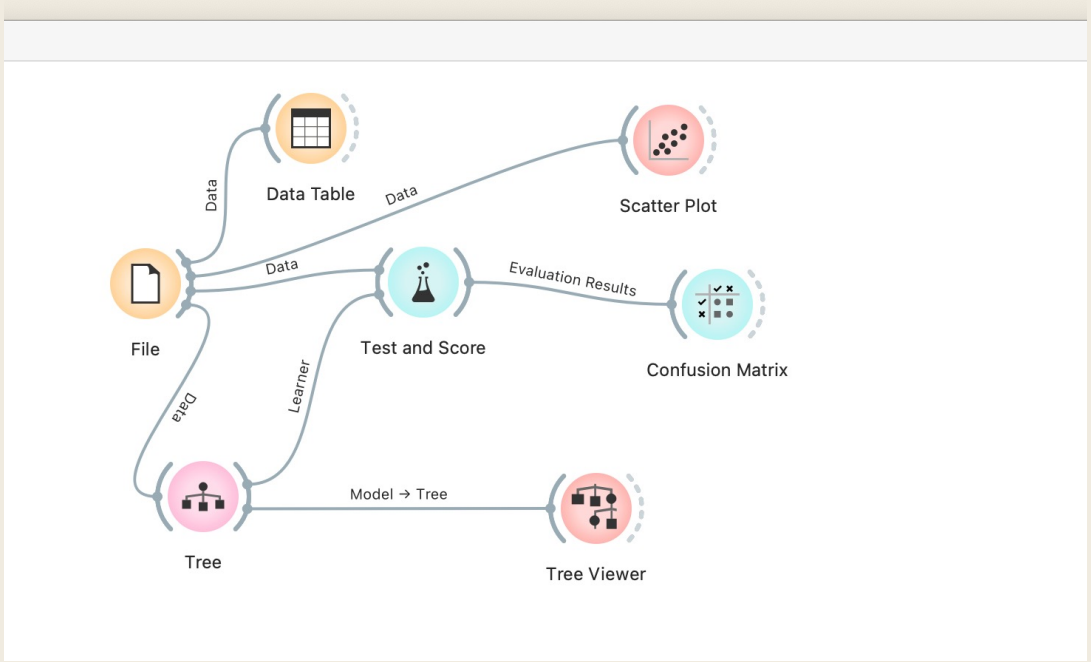
S: [3 B, 3 A]
E = 1.00

Gain (s, Sex)
= 0.940 – [(7/14)0.985 + (7/14)0.592]
=0.151

Gain (s, Cholesterol)
= 0.940 – [(8/14).811 + (6/14)1.0]
=0.048

Information gain is the information that can increase the level of certainty after splitting.

Information Gain = (Entropy before split) – (weighted entropy after split)



Decision Tree

- Iris dataset
- Test and Score –> confusion Matrix
- Tree –> Tree Viewer

Linear Models
For classification

- Linear models are a class of models that is widely used in practice.
- Linear models make a prediction using a linear function of the input features

Linear Binary classification

In this case, the prediction is made using the formula:

$$\hat{y} = w[0] * x[0] + w[1]^{*}x[1] + \cdots + w[p]^{*}x[p] + b > 0$$

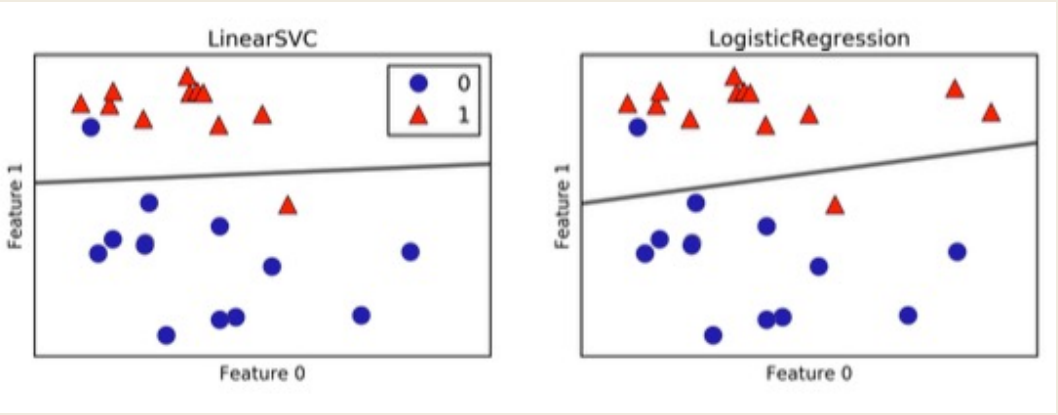
Where:
X [0,..p] are the features (columns)
W[0,..,p] and b are the parameters of the model
Y is the prediction of the model

Linear Models For classification

A linear classifier is a classifier that separates two classes using a line, a plane or a hyperplane

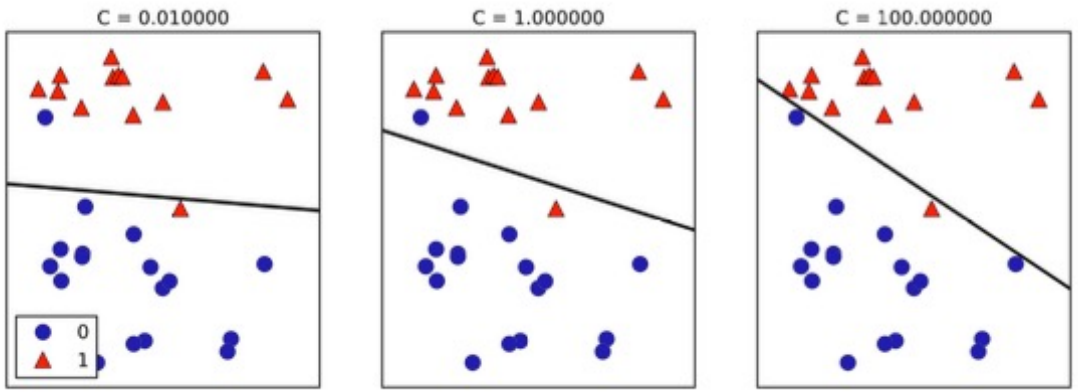
The most common models:

- Logistic Regression
- Support Vector Machine



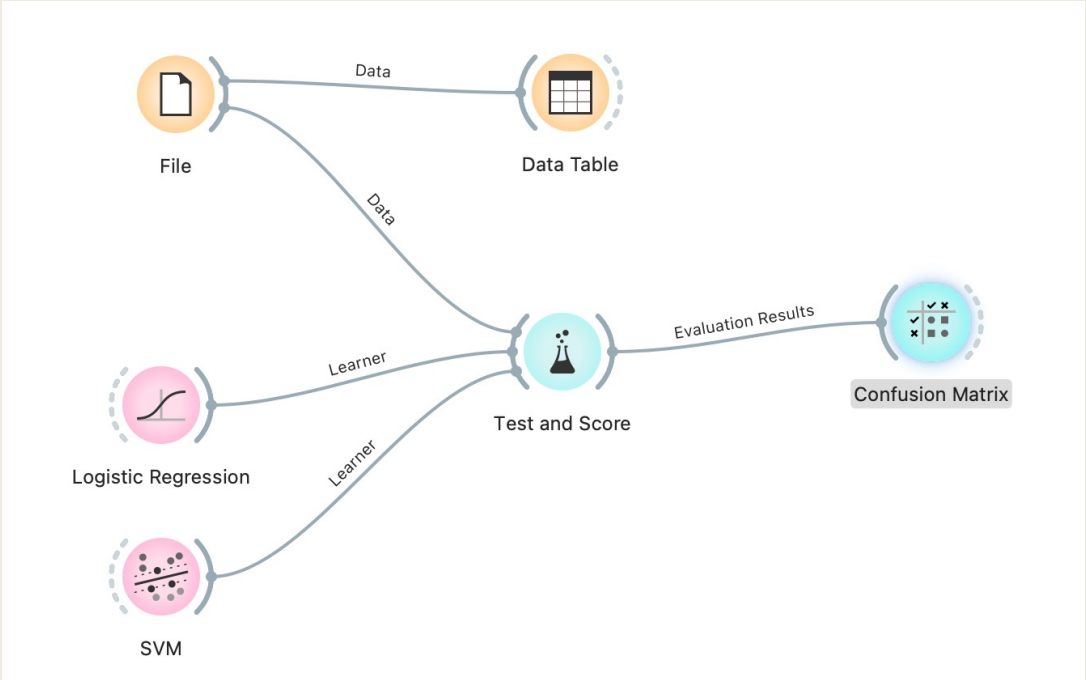
Linear models

- Linear SVC
- Logistic Regression.



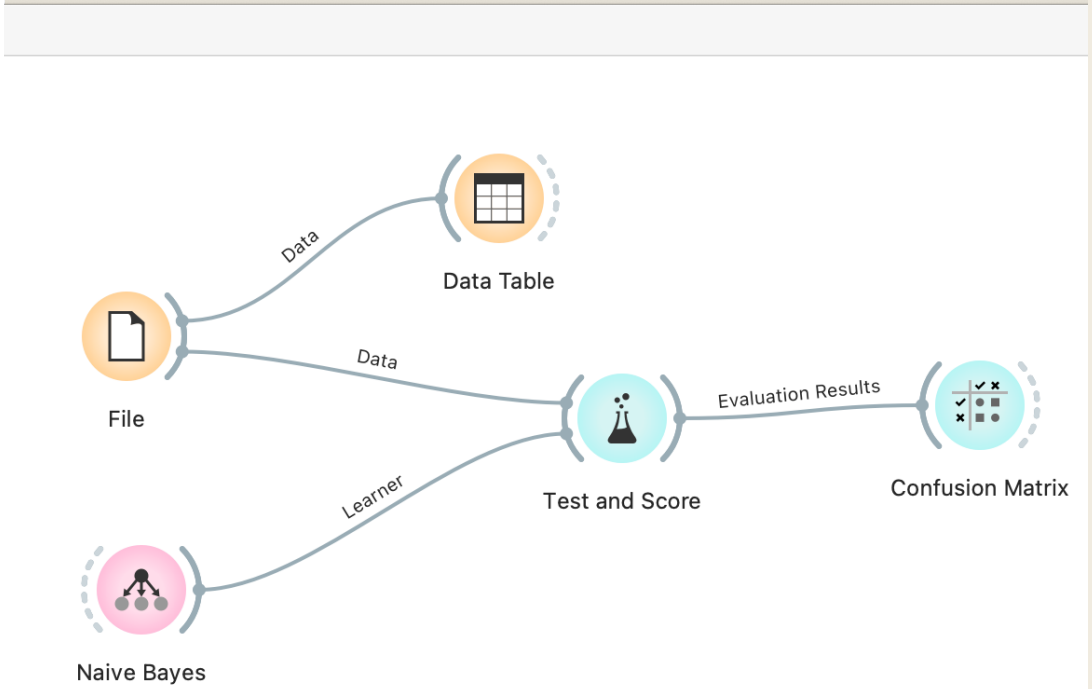
Linear models

- Studying different C values
- Studying different regularizations.



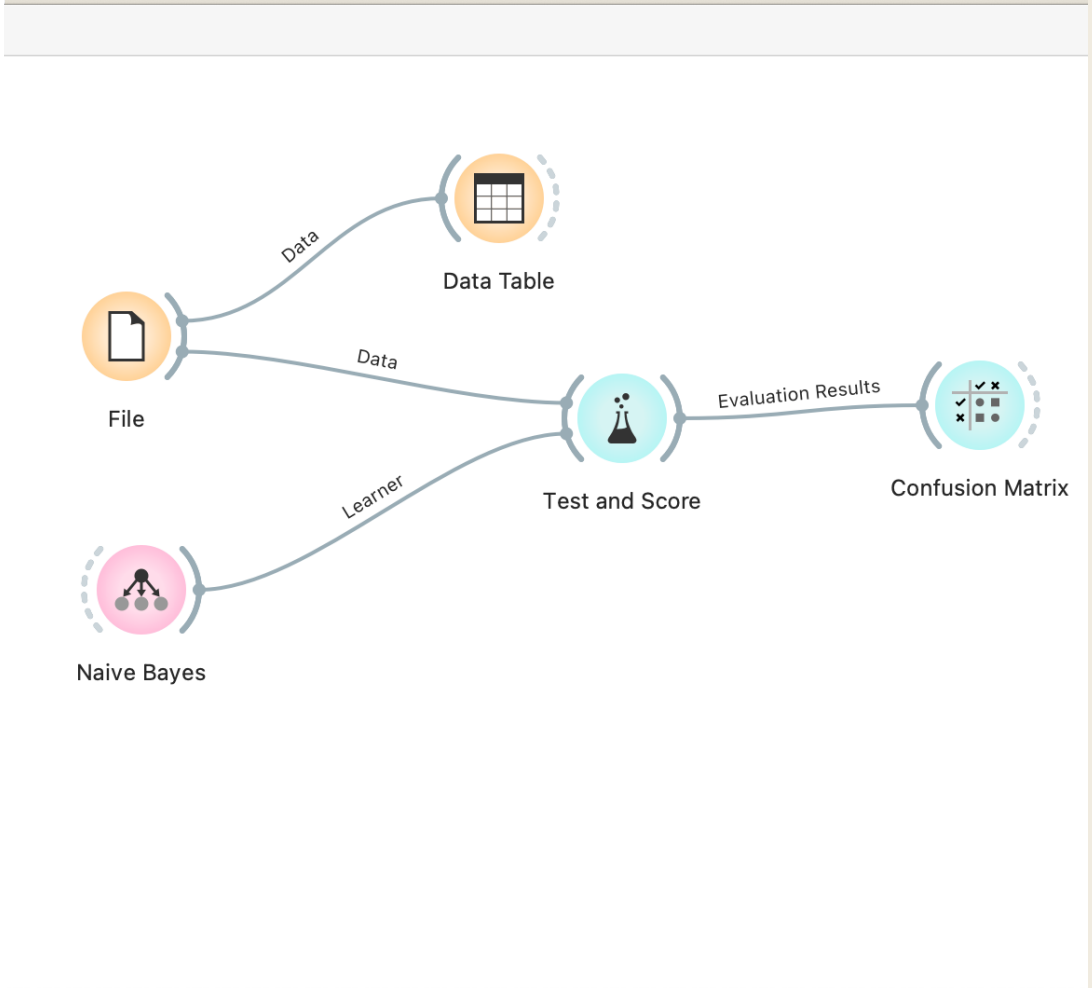
Linear Models

- Students dataset
- SVM -> Test and Score -> Confusion M.
- LR -> Test and Score -> Confusion M.
- C values
- Regularization type: L1, L2



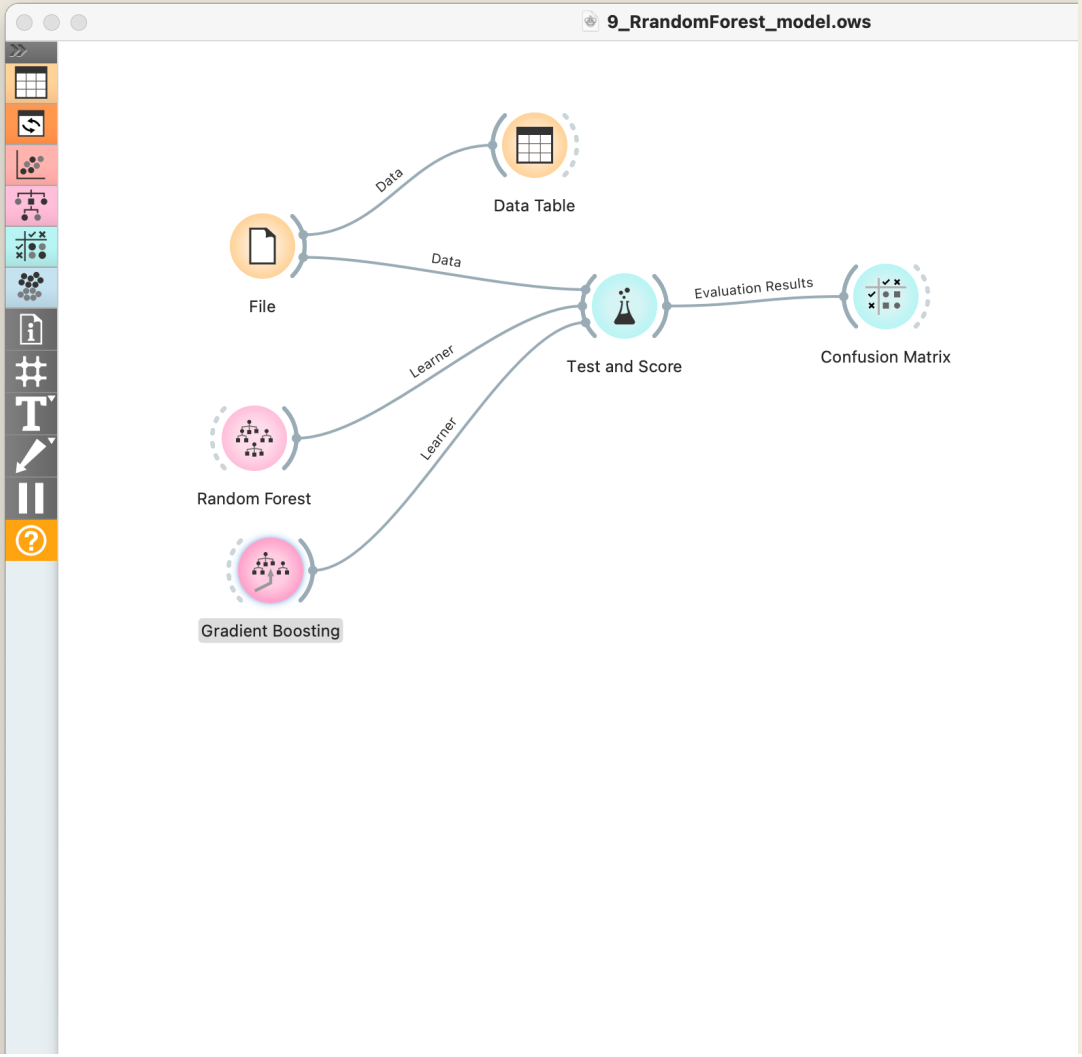
Naive Bayes models

- Naive Bayes classifiers are a family of classifiers that are quite similar to the linear models.
- However, they tend to be even faster in training.
- The price paid for this efficiency is that naive Bayes models often provide generalization performance that is slightly worse than that of linear classifiers



Naive Bayes models

- There are three kinds of naive Bayes classifiers: **GaussianNB, BernoulliNB,** and **MultinomialNB**.
- GaussianNB can be applied to any continuous data, while BernoulliNB assumes binary data and MultinomialNB assumes count data
- Naive Bayes models are great baseline models and are often used on very large datasets, where training even a linear model might take too long.

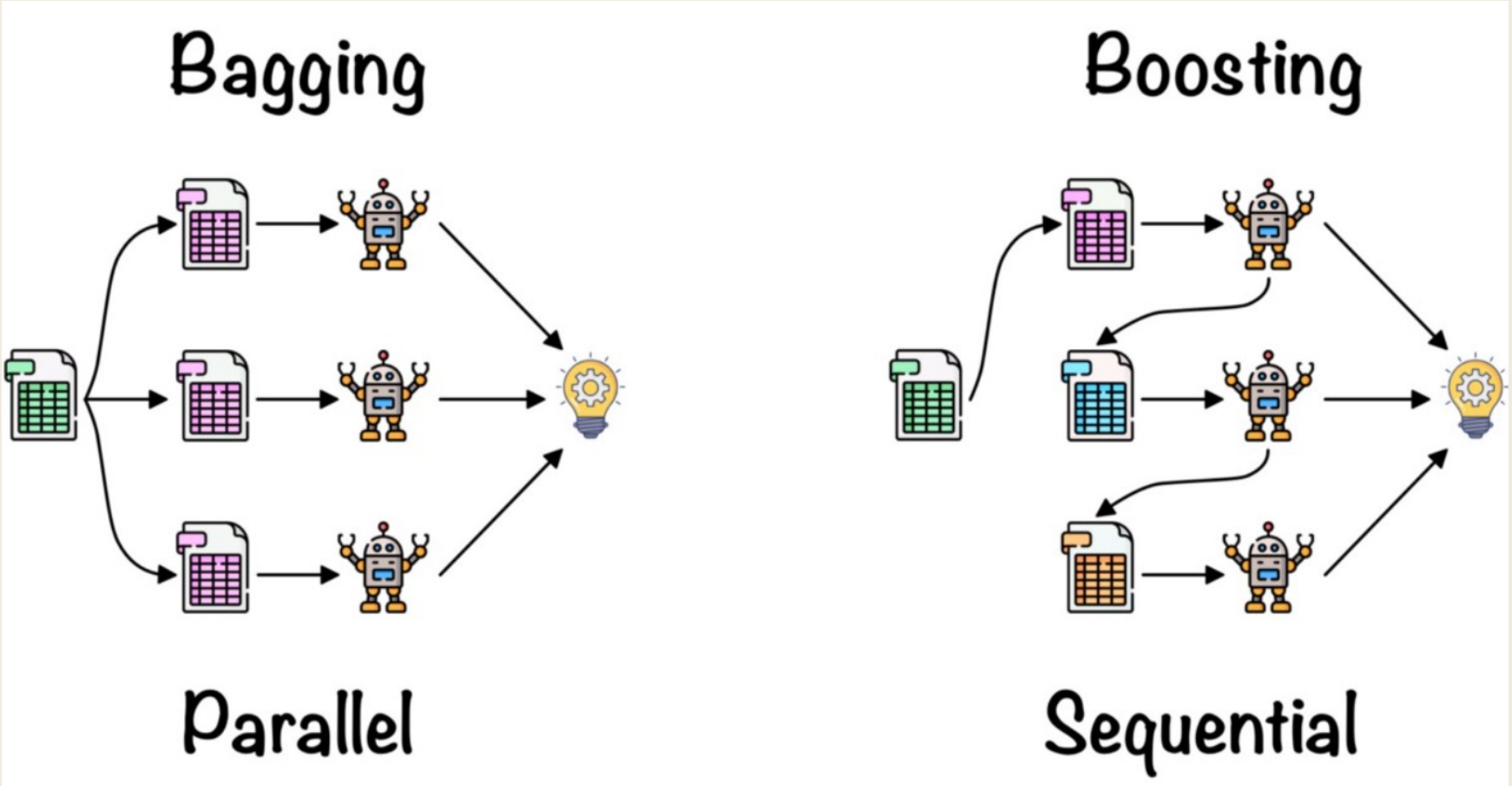


Ensembles of Decision Trees

- Ensembles are methods that combine multiple machine learning models to create more powerful models.
- There are two ensemble models that have proven to be effective, both of which use decision trees as their building blocks: **random forests** and **gradient boosted decision trees.**

Random Forest

Gradient Boosted



Random Forest

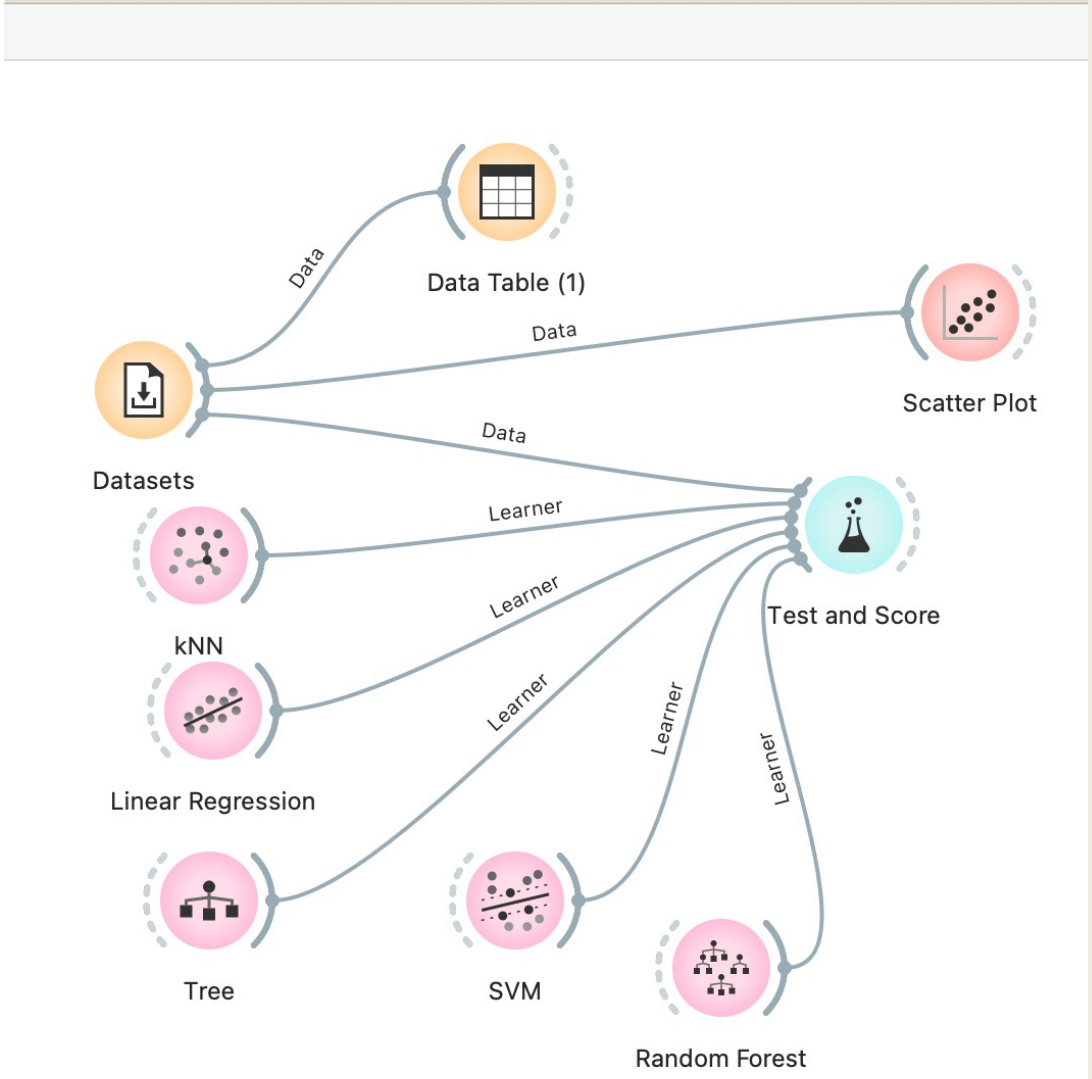
- The main drawback of decision trees is that they **tend to overfit** the training data. Random forests are one way to address this problem.
- A RF is essentially **a collection of decision trees**, where each tree is slightly different from the others. The idea behind random forests is that each tree might do a relatively good job of predicting, but will likely overfit on part of the data. If we build many trees, all of which work well and overfit in different ways, **we can reduce the amount of overfitting by averaging their results.**

Regression problems in ML

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
1	41	2	6	19	0.124	1.073	NBA001	6.3
2	47	1	26	100	4.582	8.218	NBA021	12.8
3	33	2	10	57	6.111	5.802	NBA013	20.9
4	29	2	4	19	0.681	0.516	NBA009	6.3
5	47	1	31	253	9.308	8.908	NBA008	7.2
6	40	1	23	81	0.998	7.831	NBA016	10.9
7	38	2	4	56	0.442	0.454	NBA013	1.6
8	42	3	0	64	0.279	3.945	NBA009	6.6
9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6

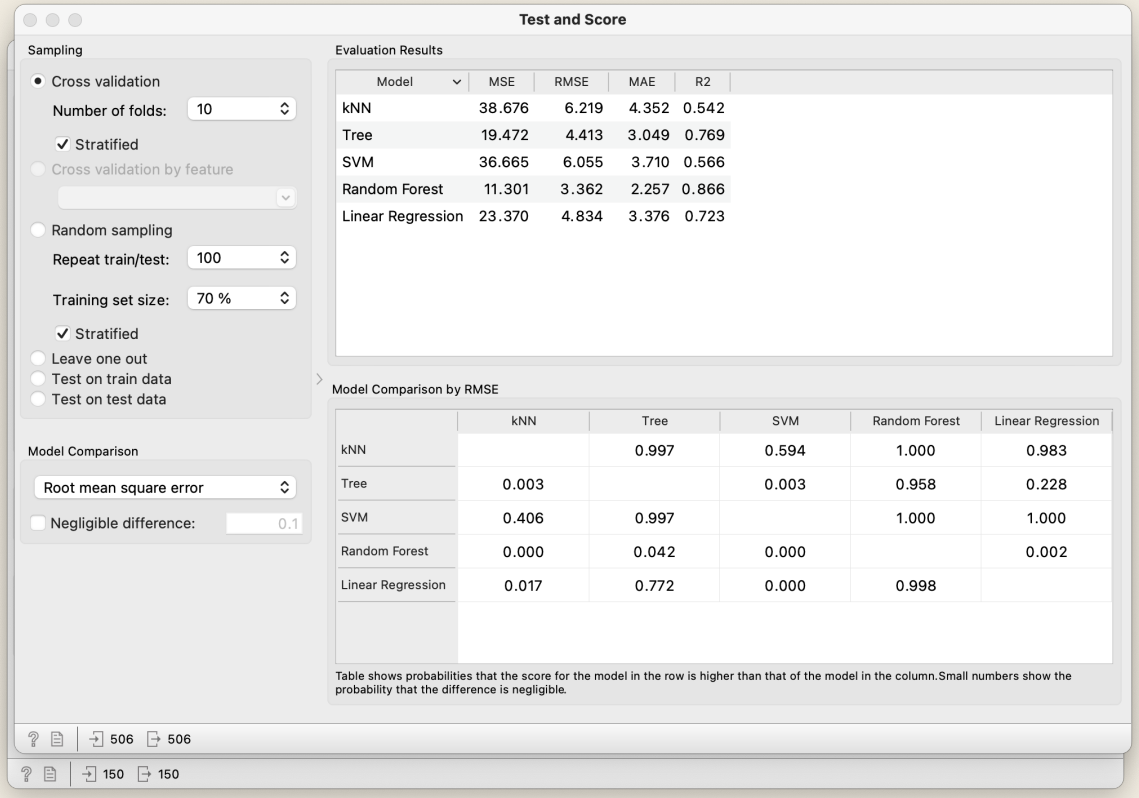
Regression

In Regression, the goal is to predict a *continuous value*



Scoring

- Abalone data (numeric Target)
- Number of goles (La Liga)
- Test & Score



Evaluation metrics in classification

- We are talking about some metrics:
  - MSE: Mean Squared Error
  - RMSE: Root Mean Squared Error
  - MAE: Mean Absolute Error
  - R^2

Error coefficients

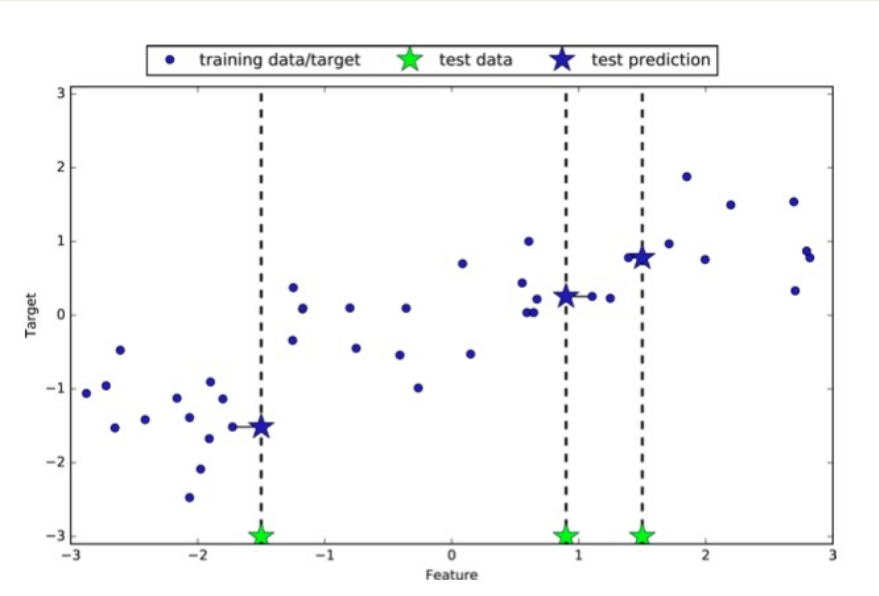
$$\mathrm{MSE} = \frac{1}{n}\sum_{i-1}^{n}\left(Y_i - \hat{Y}_i\right)^2$$

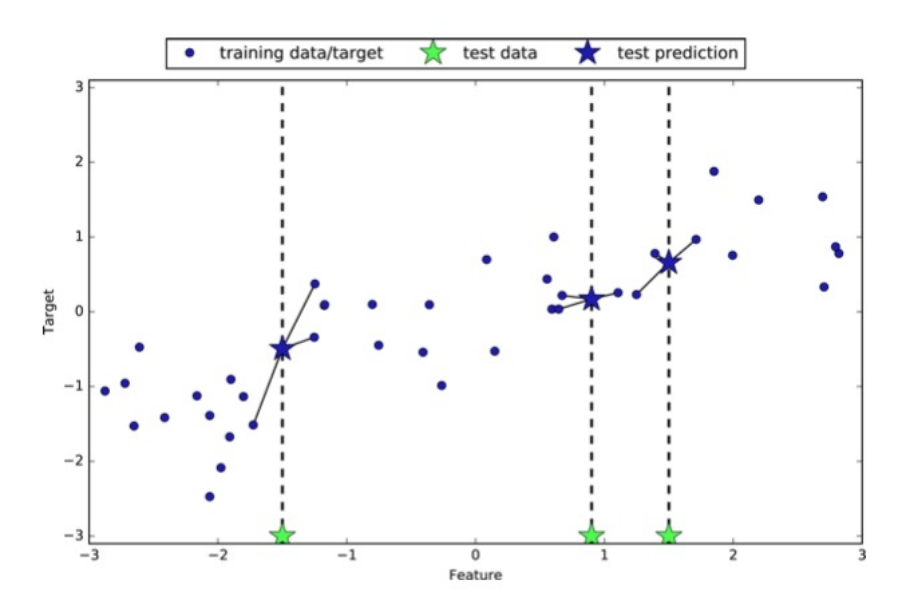
$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i-1}^{n}\left(Y_i - \hat{Y}_i\right)^2}$$

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n} = \frac{\sum_{i=1}^{n}|e_i|}{n}$$

$$R^2 = 1 - \frac{SS_{\mathrm{res}}}{SS_{\mathrm{tot}}} = 1 - \frac{\sum_i \left(y_i - \hat{y}_i\right)^2}{\sum_i (y_i - \bar{y})^2}$$

Regression Algorithms





K-Nearest Neighbors

- Building the k-NN algorithm consists only of storing the training dataset.
- To make a prediction for a new data point, the algorithm finds the closest data points in the training dataset—its "nearest neighbors."

Unsupervised Learning

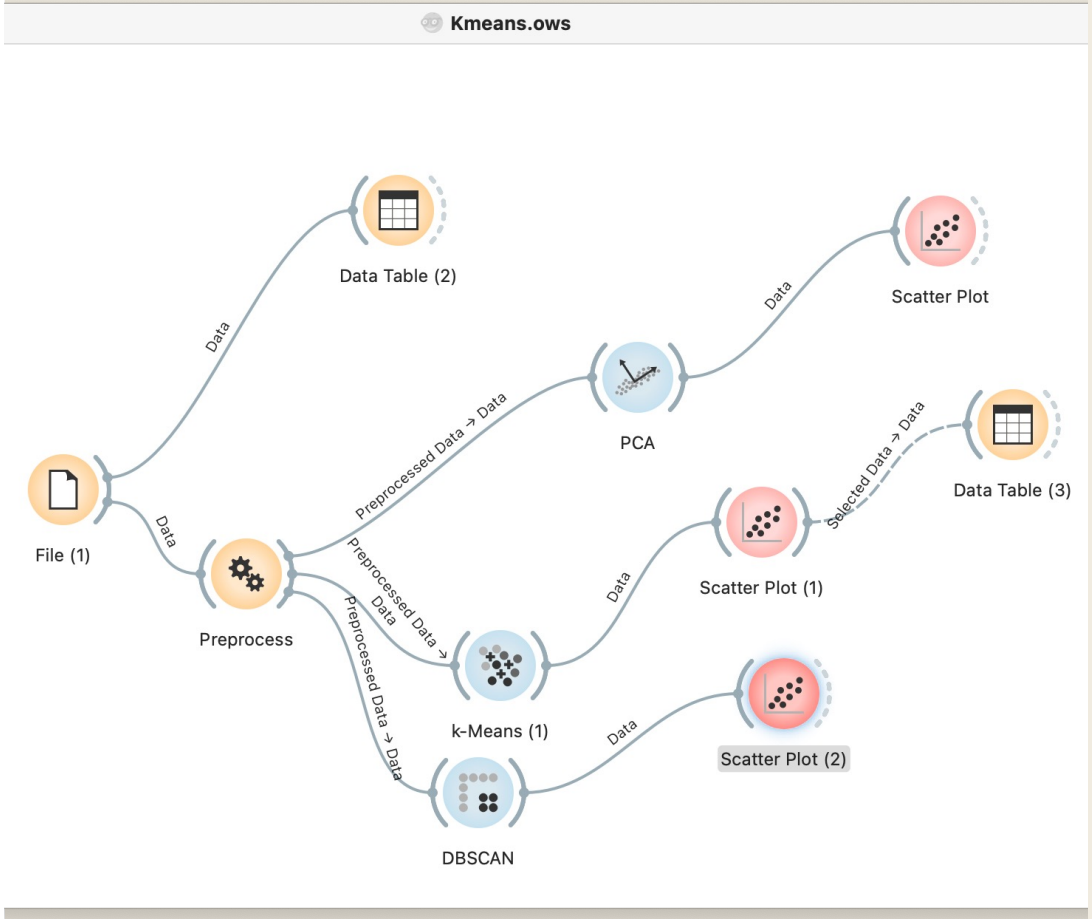
Challenges in Unsupervised Learning

A major challenge in unsupervised learning is evaluating whether the algorithm learned something useful. Unsupervised learning algorithms are usually applied to data that does not contain any label information, so we don't know what the right output should be. Therefore, **it is very hard to say whether a model "did well."**

Type of unsupervised learning

- **Unsupervised transformations** of a dataset are algorithms that create a new representation of the data which might be easier for humans or other machine learning algorithms to understand compared to the original representation of the data (dimensionality reduction)

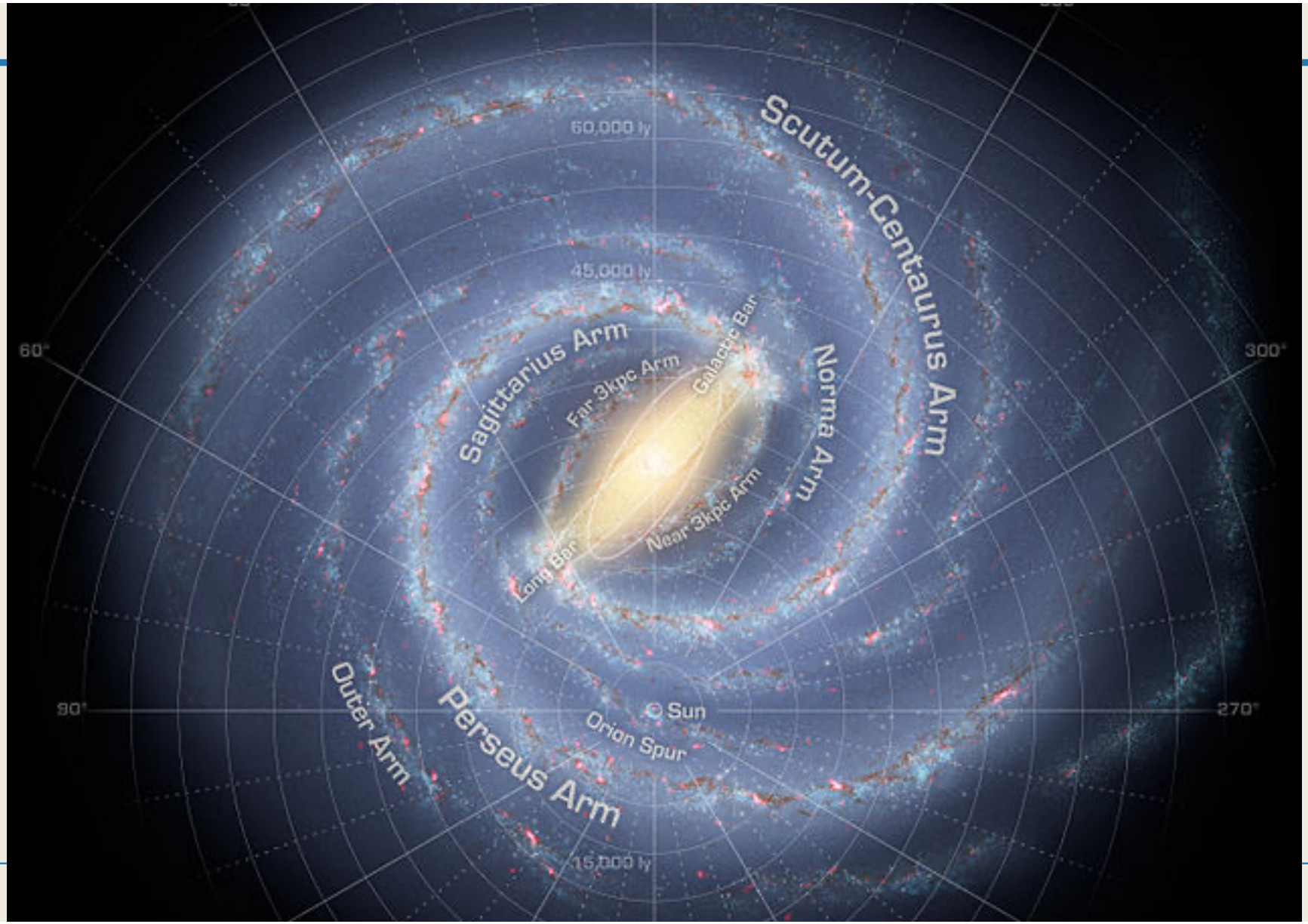
- **Clustering**, on the other hand, partition data into distinct groups of similar items

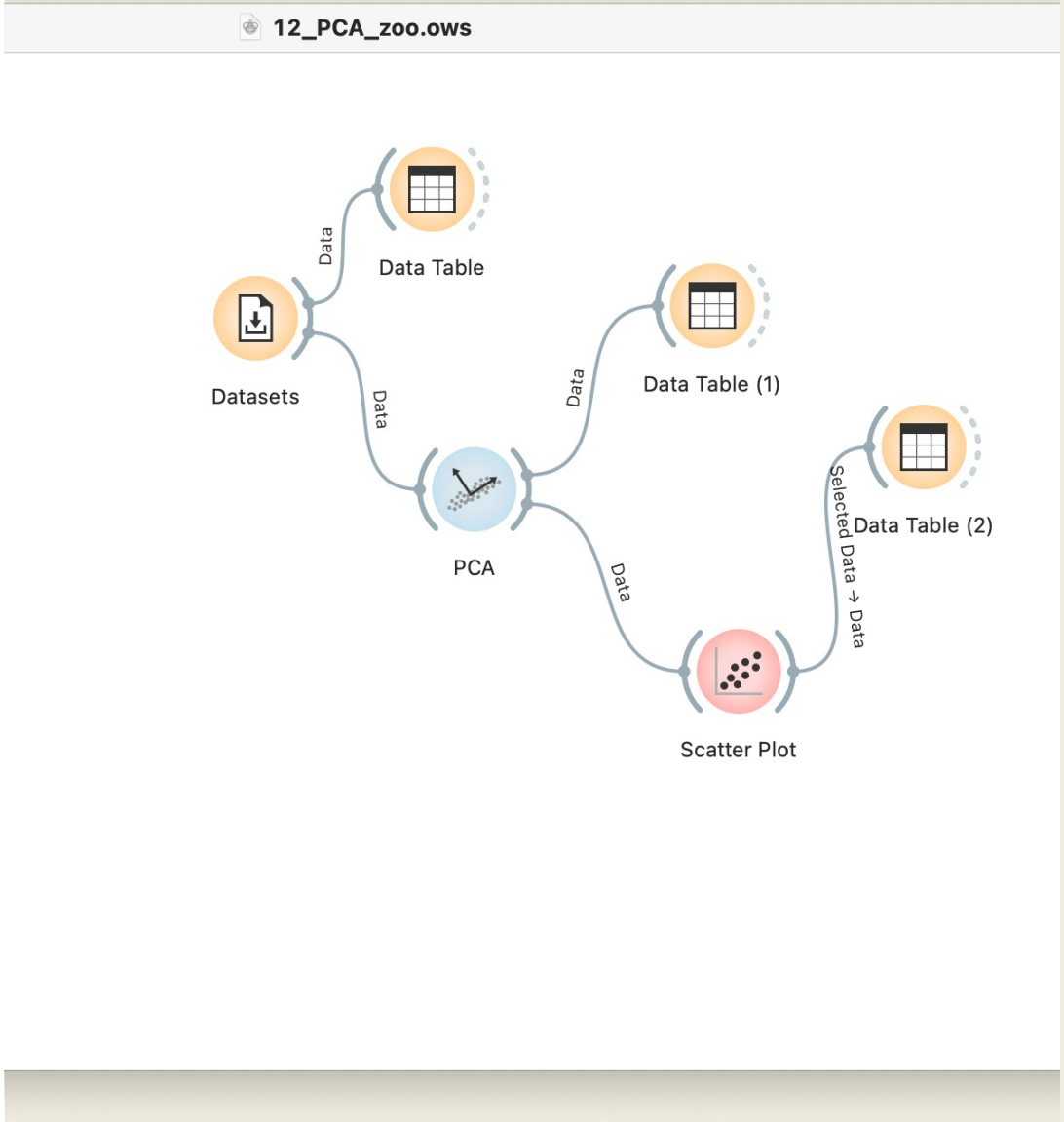


Preprocessing and Scaling

- StandardScaler
- RobustScaler
- MinMaxScaler

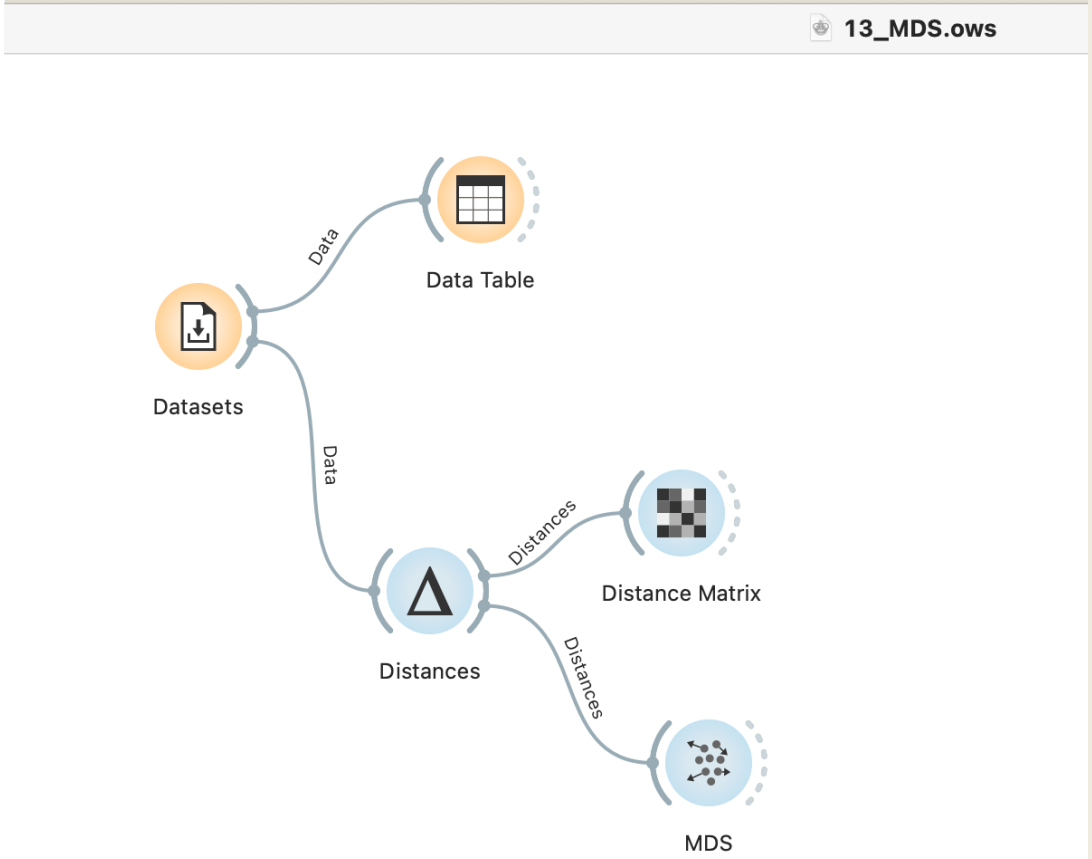

Let’s talk about stars


60,000 ly
45,000 ly
15,000 ly
60°
300°
90°
270°
Scutum-Centaurus Arm
Sagittarius Arm
Norma Arm
Perseus Arm
Outer Arm
Far 3kpc Arm
Galactic Bar
Near 3kpc Arm
Long Bar
Sun
Orion Spur



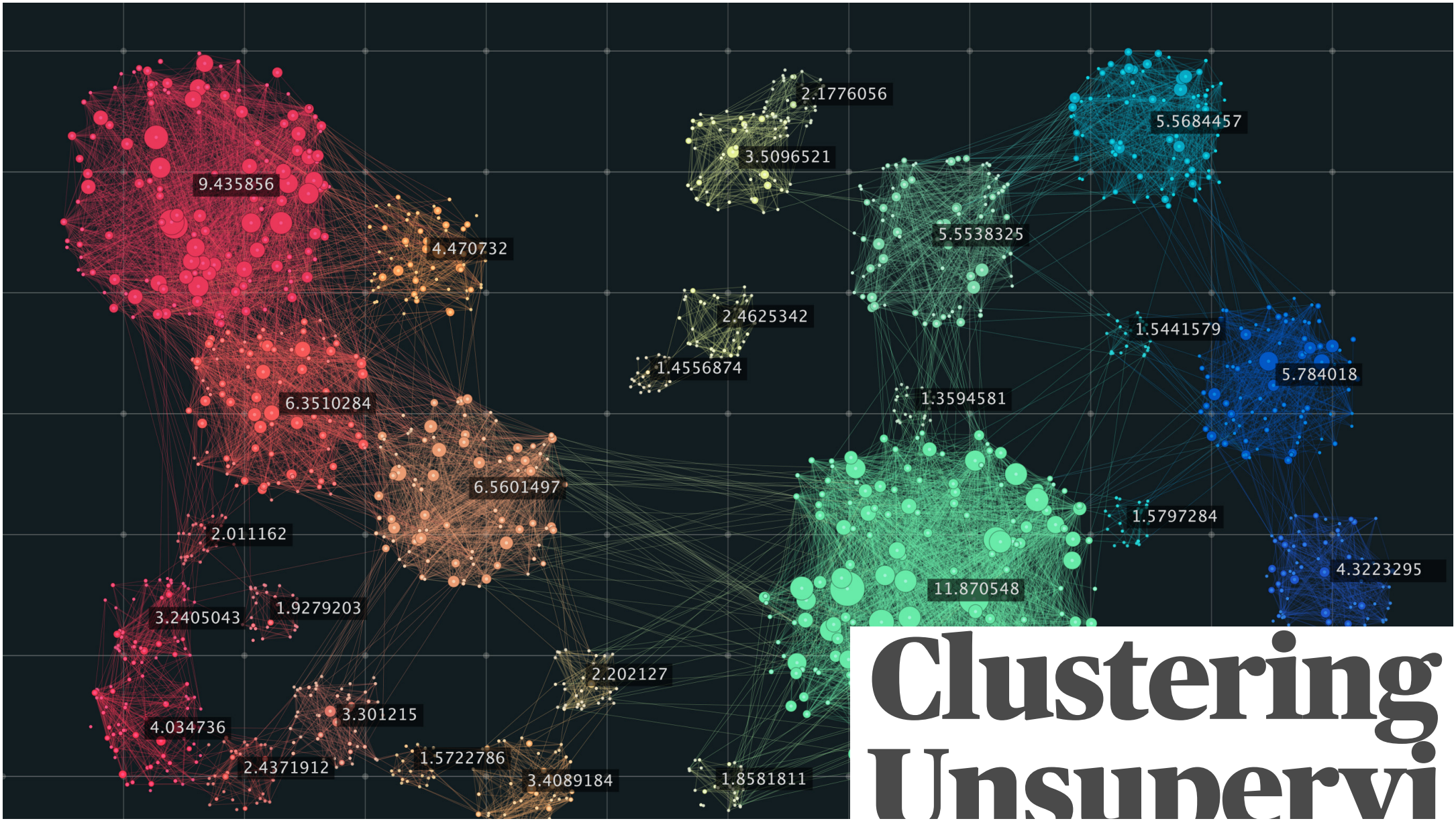
PCA

- Datasets->Zoo
- PCA
- Scatter Plot

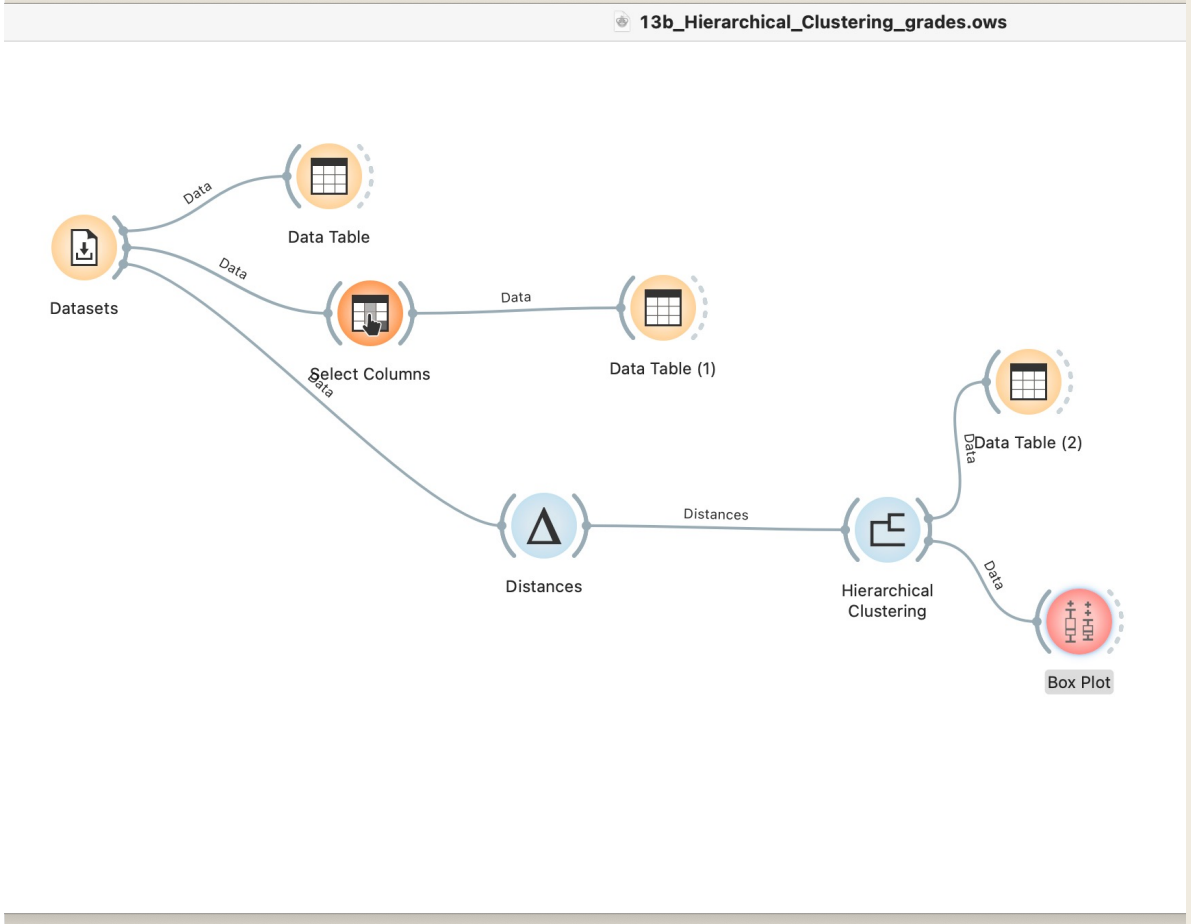


MDS and t-SNE

- Datasets->Zoo
- Distances
- MDS and t-SNE

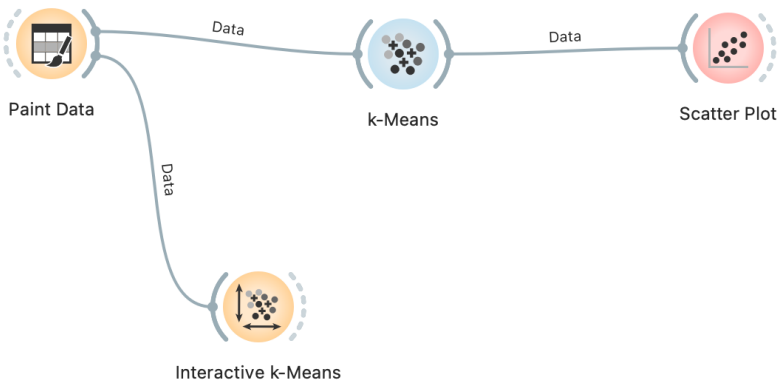


Clustering
Unsupervi



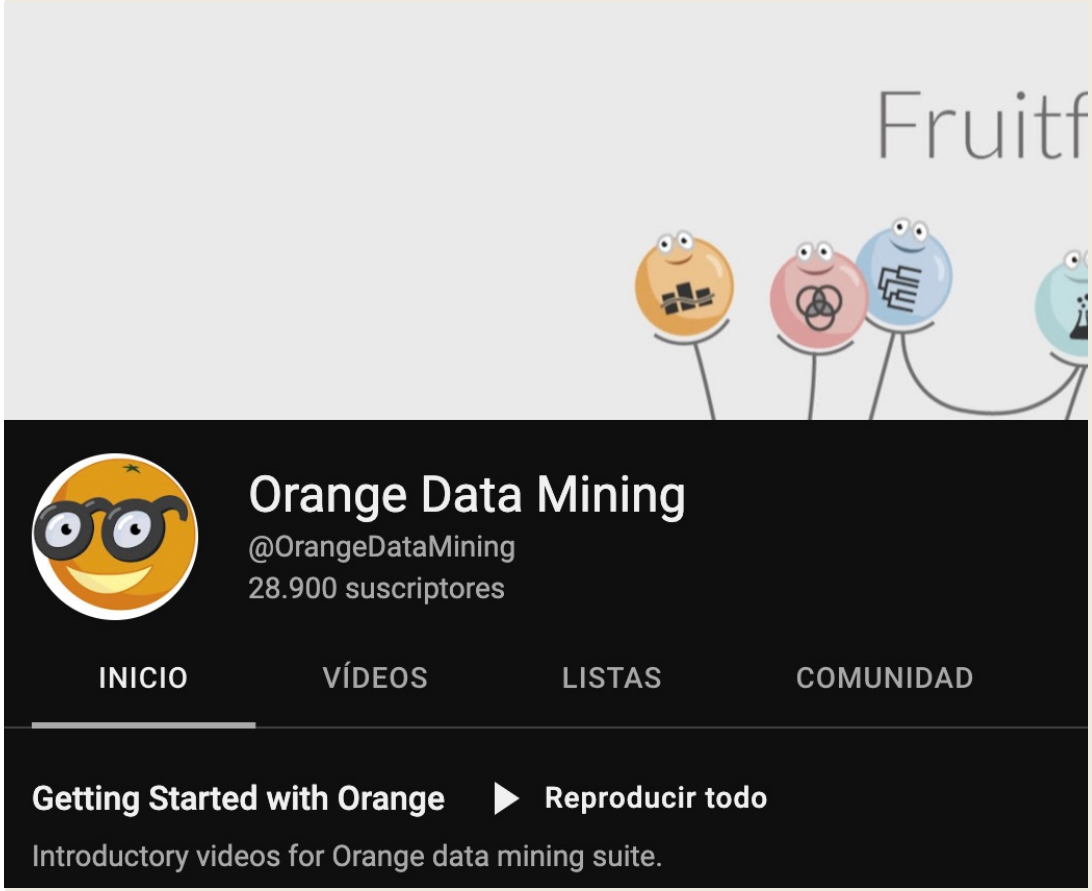
Hierarchical clustering

- Grades –> Data Table
- Distances –> hierarchical clustering
- Scatter plot
- Box plot



K-means

- Paint Data
- K-means –> Scatter plot
- Interactive K-Means
- 2 example:
- Iris Data



Bibliography

- @galeanojav
- Orange Data Mining YouTube Chanel
- The Creativity Code, Marcus du Sautoy